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1     **Medical image segmentation in a multiple**  
2     **labelers context: Application to the study of**  
3     **histopathology**

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8                     Manizales, Colombia  
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# Medical image segmentation in a multiple labelers context: Application to the study of histopathology

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27 **Segmentación de imágenes médicas en un**  
28 **contexto de múltiples anotadores:**  
29 **Aplicación al estudio de histopatologías**

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56 meaningfully to the field of early cancer detection.



## ABSTRACT

60 Breast cancer remains one of the deadliest diseases affecting women worldwide,  
61 making accurate diagnosis and treatment planning crucial for improving patient  
62 outcomes. This thesis presents a novel approach to Image Semantic Segmentation  
63 (ISS) in medical imaging, with a particular focus on histopathology images of breast  
64 cancer tissue. The work addresses the critical challenge of handling multiple  
65 annotators' labels in crowdsourcing scenarios, where annotator performance  
66 varies across the input space and image quality. The proposed solution introduces  
67 a novel loss function that implicitly infers optimal segmentation and assesses  
68 labeler reliability without requiring explicit performance inputs. This is achieved  
69 through an intermediate reliability map that enables the model to prioritize  
70 information from reliable labelers while disregarding noisy labels.

71 The methodology diverges from conventional CNN-based segmentation models by  
72 eliminating the need for direct supervision of labeler performance, thereby  
73 enhancing its generalizability and adaptability across diverse datasets and labeler  
74 cohorts. The approach combines the strengths of U-shaped deep learning  
75 architectures with a sophisticated loss function that models annotator reliability  
76 across the input space. The model's performance is evaluated on both synthetic  
77 datasets and real-world histopathology images, demonstrating superior results  
78 compared to state-of-the-art approaches in terms of segmentation accuracy and  
79 uncertainty quantification.

80 The research makes significant contributions to the field by providing a robust  
81 solution for handling inconsistent annotations in medical image segmentation,  
82 particularly valuable in histopathology where expert annotations are expensive  
83 and time-consuming to obtain. The proposed approach shows promising results in  
84 reducing annotation costs while maintaining high segmentation accuracy, making  
85 it particularly relevant for clinical applications where precise segmentation of  
86 tissue structures is crucial for diagnosis and treatment planning.

87 **Keywords:** Crowdsourcing, Multiple Annotators, Histopathology, Breast Cancer,  
88 Semantic Image Segmentation, Gaussian Processes, Deep Learning, Medical Image  
89 Analysis

91 El cáncer de mama sigue siendo una de las enfermedades más mortales en la  
92 población femenina en todo el mundo, lo que hace que el diagnóstico preciso y la  
93 planificación del tratamiento sean cruciales para mejorar las condiciones de los  
94 pacientes. Esta tesis presenta un enfoque novedoso para la Segmentación  
95 Semántica de Imágenes (ISS) en imágenes médicas, con un enfoque particular en  
96 imágenes histopatológicas de tejido de cáncer de mama. El trabajo aborda el  
97 desafío crítico de manejar las etiquetas de múltiples anotadores en escenarios de  
98 crowdsourcing, donde el rendimiento de los anotadores varía a través del espacio  
99 de entrada y la calidad de la imagen. La solución propuesta introduce una función  
100 de pérdida novedosa que infiere implícitamente la segmentación óptima y evalúa  
101 la confiabilidad de los etiquetadores sin requerir entradas explícitas sobre su  
102 rendimiento. Esto se logra a través de un mapa de confiabilidad intermedio que  
103 permite al modelo priorizar la información de etiquetadores confiables mientras  
104 descarta etiquetas ruidosas.

105 La metodología se aparta de los modelos de segmentación basados en CNN  
106 convencionales al eliminar la necesidad de supervisión directa del rendimiento de  
107 los etiquetadores, mejorando así su generalización y adaptabilidad a través de  
108 diversos conjuntos de datos y cohortes de etiquetadores. El enfoque combina las  
109 fortalezas de las arquitecturas de aprendizaje profundo en forma de U con una  
110 sofisticada función de pérdida que modela la confiabilidad de los anotadores a

111 través del espacio de entrada. El rendimiento del modelo se evalúa tanto en  
112 conjuntos de datos sintéticos como en imágenes histopatológicas del mundo real,  
113 demostrando resultados superiores en comparación con los enfoques más  
114 avanzados en términos de precisión de segmentación y cuantificación de  
115 incertidumbre.

116 La investigación hace contribuciones significativas al campo al proporcionar una  
117 solución robusta para manejar anotaciones inconsistentes en la segmentación de  
118 imágenes médicas, particularmente valiosa en histopatología donde las  
119 anotaciones de expertos son costosas y requieren mucho tiempo para obtenerlas.  
120 El enfoque propuesto muestra resultados prometedores en la reducción de costos  
121 de anotación mientras mantiene una alta precisión de segmentación, lo que lo hace  
122 particularmente relevante para aplicaciones clínicas donde la segmentación precisa  
123 de estructuras tisulares es crucial para el diagnóstico y la planificación del  
124 tratamiento.

125 **Palabras clave:** Crowdsourcing, Múltiples Anotadores, Histopatología, Cáncer de  
126 Mama, Segmentación Semántica de Imágenes, Procesos Gaussianos, Aprendizaje  
127 Profundo, Análisis de Imágenes Médicas



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327

CHAPTER

328

ONE

329

330

INTRODUCTION

331

## 1.1 Motivation

332 Since Roentgen's discovery of X-rays in 1895, medical imaging has advanced  
333 significantly, with modalities like radionuclide imaging, ultrasound, **Computed**  
334 **Tomography (CT)**, **Magnetic Resonance Imaging (MRI)**, and digital radiography  
335 emerging over the past 50 years. Modern imaging extends beyond image  
336 production to include processing, display, storage, transmission and analysis.  
337 [Zhou et al., 2021]. Other **Medical Imaging Techniques (MITs)** have arose during  
338 the last decades, some of them implying only the examination of certain pieces or  
339 tissues instead of complete patients, like histopathological images, which are  
340 images of tissue samples obtained from biopsies or surgical resections and are  
341 widely used for the diagnosis of diseases like cancer through **Whole Slide Imaging**  
342 **(WSI)** scanners [Rashmi et al., 2021].

343 Along with the advances in technologies for medical images acquisition,  
344 computational technologies on pattern recognition and artificial intelligence have

also emerged, allowing the development of **Computer-Aided Diagnosis (CAD)** systems based on machine learning algorithms. These systems aim to assist physicians in the diagnosis and treatment of diseases, by providing a second opinion or by automating the analysis of medical images. [Panayides et al., 2020]. One of the most used tasks in which machine learning technologies is being used in the universe of medical images is **Image Semantic Segmentation (ISS)**, which consists of assigning a label to each pixel in an image according to the object it belongs to. This task is crucial for the development of **CAD** systems, as it allows the identification of **Region of Interest (ROI)** in the images, which can be used to detect and classify diseases [Azad et al., 2024].

The application of Machine Learning in medical imaging has grown significantly, with key tasks including classification, segmentation, anomaly detection, super-resolution, image registration, and synthetic image generation [Brito-Pacheco et al., 2025]. Among imaging modalities, X-rays and **CT** scans are widely used for classification and anomaly detection, especially in pulmonary and oncological applications. **MRI** and ultrasound play a crucial role in segmentation and resolution enhancement, while PET/SPECT imaging is essential for anomaly detection in oncology and neurodegenerative diseases [Brito-Pacheco et al., 2025]. Histopathology is rapidly gaining prominence, particularly in segmentation and feature extraction, where AI-driven techniques aid in automated cancer diagnosis and tissue structure analysis. The integration of Deep Learning in histological image processing is revolutionizing pathology, enabling more precise and efficient diagnostics. A brief comparison of the tasks and medical image types based on recent literature review, can be seen in Figure 1-1. [Yu et al., 2025], [Brito-Pacheco et al., 2025], [Ryou et al., 2025], [Hu et al., 2025], [Elhaminia et al., 2025]

For solving the different requirements of tasks in medical images, a variety of computational techniques have been developed [Zhou et al., 2021]. Initially, these needs were covered with simple morphological filters, which implied no training process or elaborated optimization. However, as the complexity of the tasks increased, the need for more sophisticated techniques arose, leading to the



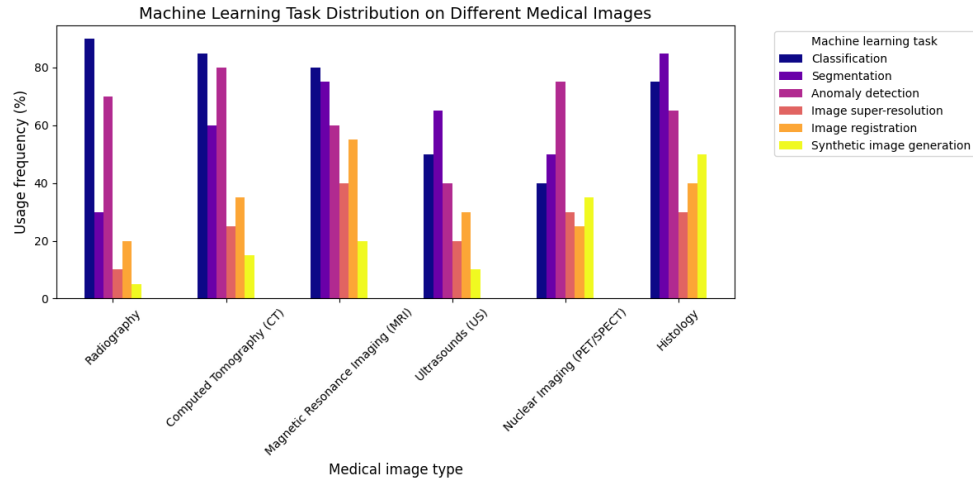


Figure 1-1 Estimation of the popularity of tasks and medical image types based on recent literature review (count of referenced terms).

375 application of advanced statistical tools and machine learning algorithms like  
 376 Support Vector Machines, Decision Trees, and SGD Neural Networks [Avanzo  
 377 et al., 2024]. The coevolution of advances in medical image acquisition,  
 378 computational power (i.e. Moore's law) and statistical/mathematical techniques  
 379 have led to a convergence for merging state of the art algorithms with medical  
 380 imaging [Shalf, 2020]. Figure 1-2 shows a brief timeline of coevolution between  
 381 some conspicuous advances in computational pattern recognition and its medical  
 382 applications in different scopes (besides medical imaging) [Avanzo et al., 2024].

383 Convolutional Neural Networks (CNN) have been widely used in Semantic  
 384 segmentation (SS) tasks, as they have outperformed traditional machine learning  
 385 algorithms in this task for both medical and non medical images [Xu et al., 2024]  
 386 [Sarvamangala and Kulkarni, 2022]. However, most CNN architectures are deep,  
 387 which imply a necessity of a large amount of data to train them. This introduces a  
 388 problem since both the acquisition and annotation of medical images are  
 389 expensive and time-consuming processes. This is especially true for ISS tasks, as  
 390 they require pixel-level annotations, which is taxing in terms of cost, time and  
 391 logistics involved [Bhalgat et al., 2018]. Other fashions face this problem through  
 392 less expensive annotation strategies like bounding boxes or anatomical landmarks

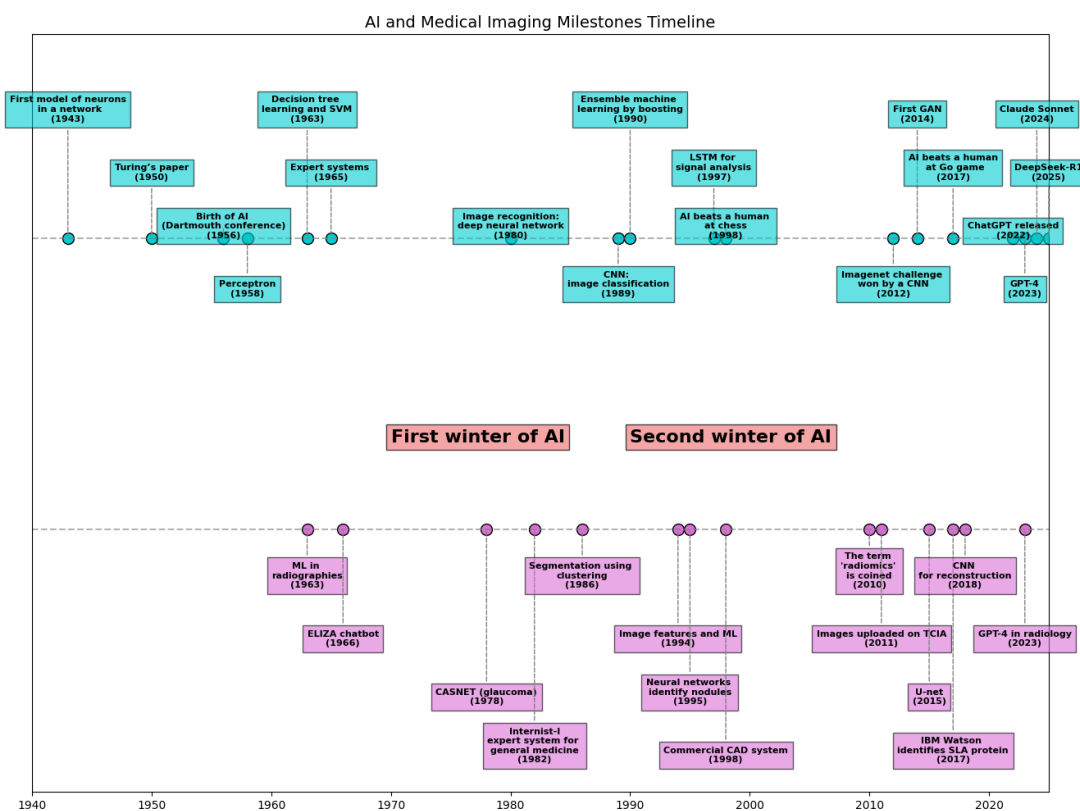


Figure 1-2 AI and machine learning in medical imaging brief timeline.

393 for being used in a semi-supervised strategy [Shah et al., 2018].

394 Many medical images datasets however, contain a high variability in class sizes  
395 and variations in colors, which is specially noticeable in histopathological images  
396 because of the usage of different staining and other factors which can affect the  
397 color of the images (see section 2.2.3). This variability can lead to a significant loss  
398 of efficiency of machine learning models when using a mixed supervision strategy,  
399 as the model can be biased towards the most common classes or colors in the  
400 dataset [Shah et al., 2018].

401 This is where other solutions arise to tackle the problem of the weak image  
402 annotation while maintaining low costs. One of these solutions is crowdsourcing  
403 strategy, which consists of having multiple annotators labeling the same image,  
404 and then combining the labels to obtain a consensus label [Lu et al., 2023]. This  
405 strategy can lead to a labeling cost reduction when different levels of expertise are  
406 combined, since the crowd may be composed of both experts and laymen, being  
407 the latter less expensive to hire [López-Pérez et al., 2023].

408 Recently, diagnosis, prognosis and treatment of cancer have heavily relied on  
409 histopathology, where tissue samples are obtained through biopsies or surgical  
410 resections and critical information that helps pathologists determine the presence  
411 and severity of malignancies [López-Pérez et al., 2024]. The segmentation of  
412 histopathological images enables precise identification of structures such as  
413 nuclei, glands, and tumors, which are essential for assessing disease progression  
414 and treatment response [Rashmi et al., 2021]. Accurate segmentation is  
415 particularly crucial in digital pathology, where whole-slide images (WSI) are  
416 analyzed using AI-powered CAD systems to support clinical decision-making  
417 [López-Pérez et al., 2024].

418 A major challenge in histopathological image segmentation arises from the  
419 variability in annotations provided by different pathologists. Unlike natural  
420 images, where object boundaries are often well-defined, histological structures  
421 may have ambiguous borders, leading to inconsistencies among annotators

[López-Pérez et al., 2023]. Because of this, crowdsourcing labeling is one of the most popular approaches, as illustrated in Figure 1-3, an example of how histopathological images are segmented by multiple experts, showing some variations in label assignment<sup>1</sup>. These discrepancies highlight the need for models that can handle annotation uncertainty effectively. Leveraging crowdsourcing strategies and machine learning techniques that infer annotator reliability can enhance segmentation performance while reducing costs.

## 1.2 Problem Statement

Throughout the development of medical technology and CAD, the task of ISS has become a crucial step in delivering precise diagnosis and treatment planning [Giri and Bhatia, 2024]. Particularly, in the area of histopathological studies, the usage of Whole Slide Images (WSI) is rather common since this method delivers high quality imaging and allows for the diagnosis of diseases like cancer [Lin et al., 2024].

ISS task consists of assigning a label to each pixel in an image according to the object it belongs to. Accurate segmentation is essential for the development of CAD systems, as it allows the identification of regions of interest (ROI) in the images, which can be used to detect and classify diseases and hence, treatment planning [Sarvamangala and Kulkarni, 2022]. However, modern computational solutions for ISS tasks involve the use of deep learning, which mostly rely large amounts of labeled data to train the models on supervised learning techniques. This means that the model is trained on a dataset with ground-truth labels, which are assumed to be correct and consistent across all samples. In practice, this assumption is often violated due to the high technical complexity of labeling these segments<sup>2</sup>.

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<sup>1</sup>obtained from a real world Triple Negative Breast Cancer (TNBC) dataset published in [López-Pérez et al., 2023]

<sup>2</sup>compared to a more trivial task like image classification on ordinary an well known classes like MNIST

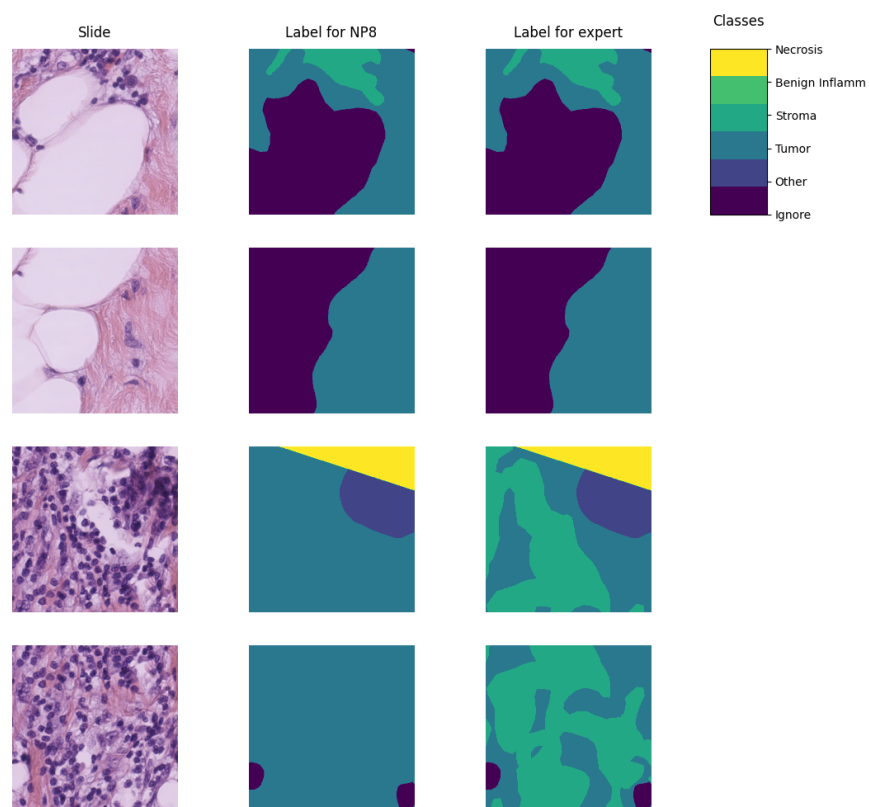


Figure 1-3 Example of a histopathological image segmented by multiple annotators, illustrating variations in label assignment.

The process of labeling medical images is often managed with the help of specialized software tools that allow the annotators to draw the regions, delivering an standard format for the labeled masks [Habis, 2024]. Despite the help of these tools, the labeling process in WSI can have high costs, as it requires long hours of work from specialized personnel. Because of cost constraints in many medical institutions, the labeling processes is often done by multiple labelers with varying levels of expertise, equalizing the cost of the labeling process. However, this strategy can lead to inconsistent labels, as the consensus between the labelers may not be exact due to the diversity in depth of knowledge and experience of the labelers [Xu et al., 2024]. These inconsistencies are mostly represented in the subsections 1.2.1 and 1.2.2.

### 1.2.1 Variability in Expertise Levels

One of the primary sources of inter-observer variability in medical image segmentation is the difference in expertise levels among annotators [López-Pérez et al., 2023]. Experienced radiologists and pathologists tend to produce highly precise annotations, whereas novice labelers may introduce systematic biases due to their limited familiarity with subtle image features. Studies have demonstrated that annotation accuracy tends to improve with experience, yet medical institutions often rely on a mix of annotators to manage costs and workload distribution [Lu et al., 2023].

The training background of annotators and institutional guidelines play a crucial role in shaping labeling practices. Different medical schools and hospitals may adopt distinct segmentation protocols, leading to inconsistencies when datasets are combined from multiple sources [López-Pérez et al., 2023]. For example, some institutions may emphasize conservative delineation of tumor boundaries, while others adopt a more inclusive approach. Such variations contribute to systematic biases in medical image datasets [Banerjee et al., 2025].

Medical images frequently contain structures with ambiguous boundaries, making segmentation inherently subjective. For instance, tumor margins in histopathological slides may not have well-defined edges, leading to variations in how different annotators delineate the regions of interest [Carmo et al., 2025]. These discrepancies arise not only from technical expertise but also from differences in perception and interpretation.

### 1.2.2 Technical Constraints and Image Quality

Technical constraints in medical imaging, such as resolution differences, noise levels, and contrast variations, can significantly impact segmentation accuracy. Lower-resolution images may obscure fine structures, leading to inconsistencies in boundary delineation [Zhou et al., 2024].

When combined with long sessions, bad images might also increase the cognitive load of the annotators, leading to fatigue and reduced precision in labeling [Kim et al., 2024]. This is particularly relevant in histopathological studies, where the staining process and tissue preparation can introduce color variations and artifacts that affect image quality, even if the same scanning equipment is used [Karthikeyan et al., 2023].

### 1.2.3 Research Question

Given the challenges posed by inconsistent labels in medical image segmentation, this work aims to address the following research question:

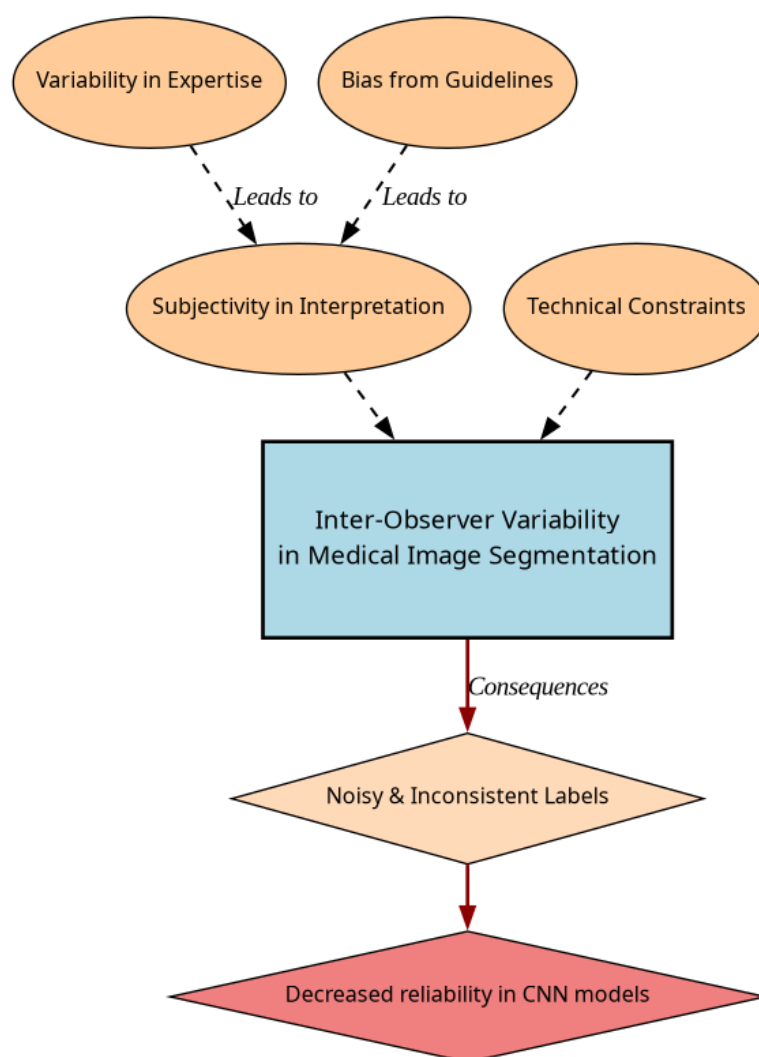


Figure 1-4 Summary diagram for problem Statement



**Research Question**

*How can we develop a learning approach for ISS tasks in medical images that can adapt to inconsistent labels without requiring explicit supervision of labeler performance, while addressing challenges related to variability in expertise levels and technical constraints, and maintaining interpretability, generalization, and computational efficiency?*

493

## 1.3 Literature Review

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Certainly, in general Machine Learning (ML) classification tasks<sup>3</sup> where multiple annotators are involved, Majority Voting (MV) is by far the simplest possible approach to implement. This concept was born multiple times and divergently in multiple fields, but it was described as relevant for ML and pattern recognition labeling for classification in [Lam and Suen, 1997], in which the approach is exposed as simple, yet powerful. The authors describe the MV as a method that can be used to improve the accuracy of classification tasks by combining the labels of multiple annotators. The method is based on the assumption that the majority vote of the annotators is more likely to be correct than the vote of a single annotator. The authors also describe the method as a straightforward way to improve the accuracy of classification tasks without the need for complex algorithms or additional data. The authors also prove this method to deliver very similar results to more complicated approaches (Bayesian, logistic regression, fuzzy integral, and neural network) in the particular task of Optical Character Recognition (OCR). Despite its simplicity, modern solutions for delivering accurate medical image segmentation models still rely on Majority Voting at some stage, like [Elnakib et al., 2020], which uses a majority voting strategy for delivering a final output based on the labels of multiple models (VGG16-Segnet, Resnet-18 and Alexnet) in CT images for Liver Tumor Segmentation, or [López-Pérez et al., 2023],

513

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<sup>3</sup>In this work, image segmentation is considered as a particular case of classification in which target classes are assigned pixel-wise.

514 which uses MV for combining noisy annotations as an additional annotator to be  
515 included in the deep learning solution. Majority voting as a technique for setting a  
516 pseudo ground truth label is a powerful approach for its simplicity in many use  
517 cases in which the target to be labeled is not tied to an expertise related task,  
518 otherwise, the assumption of equal expertise among the labelers can be a source of  
519 bias in the final label, which is not desirable in the case of highly technical  
520 annotations like medical images. In subsection 1.3.1, we will be reviewing  
521 literature which no longer assumes the naive approach of equal expertise among  
522 labelers and face the challenge of learning from inconsistent labels.

### 523 1.3.1 Facing Annotation Variability in Medical images

524 Learning from crowds approaches in general face the challenge of not having a  
525 ground truth label and hence, an intrinsic difficulty in measuring the real reliability  
526 of the labelers annotations. Some approaches assume beforehand a certain level of  
527 expertise for each labeler based on experience as an input, like in [TIAN and Zhu,  
528 2015], which introduce the concept of max margin majority voting, using the  
529 reliability vector as weights for the weights for the binary and multiclass classifier.  
530 The crowdsourcing margin is the minimal difference between the aggregated score  
531 of the potential true label and the scores for other alternative labels. Accordingly,  
532 the annotators' reliability is estimated as generating the largest margin between  
533 the potential true labels and other alternatives. The problem introduced in this  
534 approach is assuming an stationary reliability per expert across the whole input  
535 space, which is imprecise since annotators performance may change between  
536 different tasks or even between different regions of the same image.

### 537 STAPLE Mechanism

538 The Simultaneous Truth and Performance Level Estimation (STAPLE) algorithm,  
539 introduced in [Warfield et al., 2004] is a probabilistic framework that estimates a

hidden true segmentation from multiple segmentations provided by different raters. It also estimates the reliability of each rater by computing their sensitivity and specificity.

The **STAPLE** algorithm's goal is to maximize the log likelihood function:

$$(\mathbf{p}, \mathbf{q}) = \arg \max_{\mathbf{p}, \mathbf{q}} \ln f(\mathbf{D}, \mathbf{T} \mid \mathbf{p}, \mathbf{q}). \quad (1-1)$$

Where  $\mathbf{D}$  is the set of segmentations provided by the raters,  $\mathbf{T}$  is the hidden true segmentation,  $p$  is the sensitivity and  $q$  is the specificity of the raters.

This is achieved by using the Expectation-Maximization algorithm to maximize the log likelihood function in equation, which is done iteratively with step computations:

$$\begin{aligned} (p_j^{(k)}, q_j^{(k)}) = \arg \max_{p_j, q_j} & \sum_{i: D_{ij}=1} W_i^{(k-1)} \ln p_j \\ & + \sum_{i: D_{ij}=1} \left(1 - W_i^{(k-1)}\right) \ln(1 - q_j) \\ & + \sum_{i: D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\ & + \sum_{i: D_{ij}=0} \left(1 - W_i^{(k-1)}\right) \ln q_j. \end{aligned} \quad (1-2)$$

The capacity of STAPLE to accurately estimate the true segmentation, even in the presence of a majority of raters generating correlated errors, was demonstrated, which makes it theoretically a strong choice for setting a ground-truth in binary or multiclass medical **ISS** tasks.

The popularity and performance of **STAPLE** has led to its usage in modern applications medical image, 3d spatial images due to its assumption of decision

space being based on voxel-wise decisions, like the authors in [Grefve et al., 2024] which applied the algorithm on Positron Emission Tomography (PET) images. Other authors still rely heavily on STAPLE for setting a ground truth consensus for histopathological images, like [Qiu et al., 2022].

However, the STAPLE algorithm has some limitations. It assumes independent rater errors, which may not hold in practice, leading to biased estimates. STAPLE is also sensitive to low-quality annotations, potentially degrading final segmentations if the weights are not initialized correctly. The algorithm tends to over-smooth results, blurring fine details, and struggles with multi-class segmentation. Computationally, it is expensive due to its iterative EM approach. Additionally, STAPLE cannot correct systematic biases in annotations and depends on initial estimates, impacting accuracy. Lastly, the estimated performance levels lack interpretability, making it difficult to assess annotator reliability effectively.

Finally, this work contemplates STAPLE as useful for label aggregation, hence being a good support for other methods, but not that useful for providing annotations of structures on new and unlabeled images.

## U-shaped CNNs

Since the introduction of U-Net [Ronneberger et al., 2015] in 2015 for biomedical image segmentation, U-shaped CNNs have become a prevalent architecture in medical image segmentation tasks. The U-Net’s success stems from its ability to capture both global and local information through its contracting and expanding paths, making it particularly effective for complex and heterogeneous structures, even with limited annotated data. This architecture has been successfully applied to various medical image segmentation tasks, including organ segmentation, tumor segmentation, and brain structure segmentation.

The U-Net architecture consists of a symmetric encoder-decoder structure with skip connections. The encoder path progressively reduces spatial dimensions

while increasing feature channels through a series of convolutional and max-pooling layers, capturing high-level semantic information. The decoder path uses transposed convolutions to gradually recover spatial resolution while reducing feature channels. Skip connections between corresponding encoder and decoder layers preserve fine-grained details by concatenating high-resolution features from the encoder with upsampled features in the decoder, enabling precise localization of structures.

### U-Net based approaches

In [López-Pérez et al., 2024] two networks are trained for delivering a final segmentation. One network is trained to estimate the annotators reliability and another one is trained to segment the image. The first network is a deep neural network that takes as input features of image and the labelers id encoded as one-hot and outputs a reliability map across the image feature space. This map is then used to weight the contribution of each annotator to the final segmentation. The second network is the U-Net used for segmentation.

In this approach, it is assumed that the images are labeled for at least one labeler and not all of them, which is closer to a real world scenario, in which it is common to have images with variability in the amount of annotations, per patch. Hence, the input data can be modeled as:

$$\mathcal{D} = (\mathbf{X}, \tilde{\mathbf{Y}}) = \{(\mathbf{x}_n, \tilde{\mathbf{y}}_n^r) : n = 1, \dots, N; r \in R_n\}, \quad (1-3)$$

Where every  $\mathbf{x}_n$  is an input patch from a ROI in one WSI,  $\tilde{\mathbf{y}}_n$  is the noisy annotation from the  $r$  labeler,  $N$  is the number of patches in the dataset and  $R_n \subset \{1, \dots, R\}$  is the set of labelers that annotated the image  $\mathbf{x}_n$ .

The authors then assume the annotator network to deliver a reliability map  $\{\hat{\mathbf{A}}_\phi^{(r)}(\mathbf{x})\}_{r \in R_n}$  with different dimensions:

- 606 • CR global: a single reliability vector per labeler with dimensions  $C$  which  
607 represent global reliability of the labeler across all input space.
- 608 • CR image: a single reliability vector per image per labeler with dimensions  $C$   
609 which represent local reliability of the labeler across the image.
- 610 • CR pixel: a reliability matrix per image per labeler, with dimensions  $C$  which  
611 represent local reliability of the labeler across all the pixels in the image.

612 These differences in dimensions are determined by the feature extraction space  
613 from segmentation network which feed the input of the annotator network, which  
614 the authors vary for experimentation purposes.

615 Being  $\mathbf{p}_\theta(\mathbf{x}_n)$  the estimation of the latent (ground truth) segmentation delivered by  
616 the segmentation UNet network, thus, the estimated segmentation probability  
617 mask for each annotator is given by the product:

$$\mathbf{p}_{\theta,\phi}^{(r)}(\mathbf{x}_n) := \mathbf{A}_\phi^{(r)}(\mathbf{x}) \odot \mathbf{p}_\theta(\mathbf{x}_n), \quad (1-4)$$

618 where  $\odot$  is the element-wise product and  $\phi$  and  $\theta$  are the parameters of the  
619 annotator network and the segmentation UNet network, respectively, being the  
620 latter initialized with a ResNet34 backbone pre-trained on ImageNet.

621 The authors propose a loss function involving cross-entropy and a trace based  
622 regularization on the reliability map, originally proposed in [Zhang et al., 2020]  
623 which combined, looks like:

$$\mathcal{L}(\theta, \phi) := \sum_{n=1}^N \sum_{r=1}^R \mathbb{I}(\tilde{\mathbf{y}}_n^{(r)} \in R_n) \cdot \left[ \text{CE} \left( \mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \cdot \mathbf{p}_\theta(\mathbf{x}_n), \tilde{\mathbf{y}}_n^{(r)} \right) + \lambda \cdot \text{tr} \left( \mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \right) \right] \quad (1-5)$$

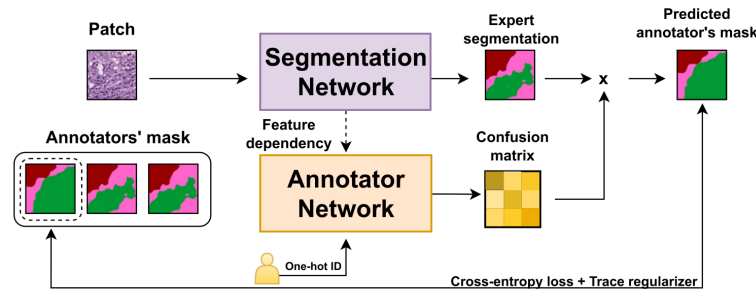


Figure 1-5 Proposed framework for the approach in [López-Pérez et al., 2024].

Being  $\mathbb{I}$  the indicator function, CE the cross-entropy loss, and  $\lambda$  the regularization parameter.

When evaluated on a Triple Negative Breast Cancer dataset, this approach achieves a Dice coefficient of 0.7827, outperforming STAPLE (0.7039) and matching expert-supervised performance (0.7723). The CR image reliability modeling proved most effective, as CR pixel, while potentially offering finer-grained reliability estimation, requires significantly more training data.

Despite the decent performance of the approach, solving the problem of multiple labelers with two networks can be overwhelming for the optimization process, requiring large amounts of annotated data to properly codify the annotators spatial reliabilities, which could be managed by a single model with an appropriate loss function.

## Bayesian models

Bayesian approaches are a good choice for handling label noise and uncertainty in the labelers. In [Julián and Álvarez Meza Andrés Marino, 2023] the authors propose a novel approach from Gaussian Processes to model the relationship between the annotators' reliability and the input data, while also preserving the interdependencies among the annotators. This is achieved by introducing Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA), a

framework based on the well known **Chained Gaussian Processes (CGP)**. CGP on itself cannot consider inter-annotator dependencies, thus, the authors introduce the **Correlated Chained Gaussian Processes (CCGP)** to model correlations between the GP latent functions, which are supposed to be generated from a **Semi-Parametric Latent Factor Model (SLFM)**:

$$f_j(\mathbf{x}_n) = \sum_{q=1}^Q w_{j,q} \mu_q(\mathbf{x}_n), \quad (1-6)$$

where  $f_j : \mathcal{X} \rightarrow \mathbb{R}$  is a **Latent Function (LF)**,  $\mu_q(\cdot) \sim \mathcal{GP}(0, k_q(\cdot, \cdot))$  with  $k_q : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$  being a kernel function, and  $w_{j,q} \in \mathbb{R}$  is a combination coefficient ( $Q \in \mathbb{N}$ ). This leads to a joint distribution of the form:

$$p(\mathbf{y}, \hat{\mathbf{f}}, u | \mathbf{X}) = p(\mathbf{y} | \boldsymbol{\theta}) \prod_{j=1}^J p(\mathbf{f}_j | \mathbf{u}) p(\mathbf{u}), \quad (1-7)$$

where  $\mathbf{y}$  is the vector of noisy labels,  $\hat{\mathbf{f}}$  is the vector of latent functions,  $u$  represents the inducing points, and  $\mathbf{X}$  is the input data.

Combined with inducing-variables based methods for sparse GP approximations, and maximizing an **Evidence Lower Bound (ELBO)** for the estimation of the variational parameters, the authors reach a model whose variational expectations are not analytically tractable, and hence, the authors derive a Gaussian-Hermite quadrature approach.

Finally, the authors extend this approach for being applied to classification and regression, reaching the only known approach to involve chained gaussian processes in multiple annotators classification and regression tasks while



661 preserving the interdependencies among the annotators, and also outperforming  
 662 GPC-MV<sup>4</sup>, MA-LFC-C<sup>5</sup>, MA-DGRL<sup>6</sup>, MA-GPC<sup>7</sup>, MA-GPCV<sup>8</sup>, MA-DL<sup>9</sup>, KAAR<sup>10</sup>.

663 CCGPMA on itself proposes a good approach for handling label noise and  
 664 uncertainty in the labelers for regression and classification tasks, while also  
 665 preserving the interdependencies among the annotators, however, it does not face  
 666 the image segmentation problem, which is the main focus of this works, however,  
 667 it does not face the image segmentation problem, which is the main focus of this  
 668 work. Besides, handling so many latent functions during the optimization process  
 669 is computationally expensive, making it on itself infeasible for large and high  
 670 resolution datasets.

### 671 1.3.2 Facing Noisy Annotations and Low-Quality Data

672 The problem of low-quality data and noisy annotations has been tackled with  
 673 various strategies. One such approach is the use of deep learning models that  
 674 incorporate loss functions designed to mitigate the effects of unreliable labels.  
 675 Traditional methods such as Majority Voting (MV) or Expectation-Maximization  
 676 (EM) have been widely used for aggregating multiple annotators' inputs. However,  
 677 they assume a homogeneous reliability of annotators, which may not hold in  
 678 real-world scenarios.

---

<sup>4</sup>A GPC using the MV of the labels as the ground truth.

<sup>5</sup>A LRC with constant parameters across the input space.

<sup>6</sup>A multi-labeler approach that considers as latent variables the annotator performance.

<sup>7</sup>A multi-labeler GPC, which is an extension of MA-LFC.

<sup>8</sup>An extension of MA-GPC that includes variational inference and priors over the labelers' parameters.

<sup>9</sup>A Crowd Layer for DL, where the annotators' parameters are constant across the input space.

<sup>10</sup>A kernel-based approach that employs a convex combination of classifiers and codes labelers dependencies.

## Loss functions in deep learning models

Loss functions are fundamental components in deep learning models that quantify how well a model's predictions match the ground truth. They serve as the objective function that guides the learning process by measuring the discrepancy between predicted and actual values. In classification tasks, the most common loss functions are Cross-Entropy (CE) and Mean Absolute Error (MAE) [Zhang and Sabuncu, 2018]. CE is particularly effective for classification as it heavily penalizes confident but wrong predictions, though it can be sensitive to noisy labels. MAE, on the other hand, is more robust to outliers and assigns equal weights to all mistakes, but typically requires more training iterations. For image segmentation tasks, specialized loss functions have been developed to handle the unique challenges of pixel-wise classification. The Dice loss, which measures the overlap between predicted and ground truth regions, is widely used in medical image segmentation [Zhao et al., 2020]. More recently, the Generalized Cross Entropy (GCE) loss has emerged as a robust alternative that combines the benefits of both CE and MAE, allowing for better handling of noisy labels through a tunable parameter that controls sensitivity to outliers. However, to the best of our knowledge, none of the current loss functions approaches have focused on the multiple labelers scenario, neither on the per labeler spatial reliability estimation.

## Generalized Cross-Entropy for multiple annotators classification

A more recent approach was proposed by [Triana-Martinez et al., 2023], introducing a Generalized Cross-Entropy-based Chained Deep Learning (GCECDL) framework. This method addresses the limitations of traditional label aggregation techniques by modeling each annotator's reliability as a function of the input data. The approach effectively mitigates the impact of noisy labels by using a noise-robust loss function, balancing Mean Absolute Error (MAE) and Categorical Cross-Entropy (CE). Unlike prior approaches, GCECDL accounts for the dependencies among annotators while encoding their non-stationary behavior

across different data samples. Their experiments on multiple datasets demonstrated superior predictive performance compared to state-of-the-art methods, particularly in cases where annotations were highly inconsistent.

The strategy of the authors effectively unlocks the potential of ML models to handle low-quality data and noisy annotations, but it is bounded to classifications tasks only, not being by itself applicable to segmentation tasks. The TGCE equation for handling multiple annotators is defined as:

$$\text{TGCE}(\mathbf{y}, f(\mathbf{x}); \tilde{\lambda}_x, \tilde{C}) = \tilde{\lambda}_x \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q} + (1 - \tilde{\lambda}_x) \frac{1 - (\tilde{C})^q}{q}, \quad (1-8)$$

where  $\tilde{\lambda}_x$  represents the annotator reliability,  $\tilde{C}$  is a constant,  $q$  is a parameter that controls the balance between MAE and CE behavior,  $\mathbf{y}$  is the annotation vector, and  $f(\mathbf{x})$  is the model prediction. This approach is more deeply discussed in chapter 4.

## 1.4 Aims

### 1.4.1 General Aim

The main purpose of this work is to develop a novel approach for ISS tasks in medical images, which can adaptively infer the best possible segmentation without needing separate inputs about the labeler performance. This approach is expected to properly offer a much more interpretable reliability space per labeler than other solutions in the literature, correctly facing the labeler performance inconsistency across the annotators space and the variability of images quality.

**Table 1-1** Summary of state-of-the-art approaches for handling multiple annotators in medical image segmentation

| Approach                  | Key Characteristics                                                        | Advantages                                                                                                           | Limitations                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Majority Voting (MV)      | Simple aggregation of multiple annotators' labels                          | - Simple to implement - No complex algorithms required - Proven effective in many cases                              | - Assumes equal expertise - Sensitive to correlated errors - May introduce bias in expert tasks                              |
| STAPLE                    | Probabilistic framework estimating true segmentation and rater reliability | - Handles binary and multiclass segmentation - Estimates rater sensitivity/specificity - Robust to correlated errors | - Computationally expensive - Sensitive to low-quality annotations - Tends to over-smooth results - Limited interpretability |
| U-Net based approaches    | Deep Learning architecture with encoder-decoder structure                  | - Captures both global and local information - Effective for complex structures - Works well with limited data       | - Requires large amounts of training data - Complex optimization process - May need specialized loss functions               |
| Bayesian (CCGPMA)         | Models Gaussian Processes modeling annotator reliability                   | - Preserves annotator interdependencies - Handles label noise effectively - Works for classification and regression  | - Computationally expensive - Not directly applicable to segmentation - Complex optimization process                         |
| Generalized Entropy (GCE) | Cross-Loss function balancing MAE and CE                                   | - Robust to noisy labels - Accounts for non-stationary behavior - Handles annotator dependencies                     | - Limited to classification tasks - Requires parameter tuning - May need adaptation for segmentation                         |

## 1.4.2 Specific Aims

- To develop a novel loss function for ISS tasks in medical images, capable of inferring the best possible segmentation without needing separate inputs about the labeler performance.
- Introducing a tensor map which codifies the reliability of each labeler, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space.
- To develop and test a deep learning model for ISS tasks in medical images, which can learn from inconsistent labels and improve the segmentation performance compared to other solutions in state of the art.

## 1.5 Outline and Contributions

Building upon the considerations discussed in Section 1.3, this thesis introduces a novel approach to ISS in medical imaging. The core of this work is a model designed to adapt its learning process to varying labeler performance. This is achieved through a unique loss function that infers optimal segmentation and assesses labeler reliability without requiring explicit performance inputs. This loss function implicitly weighs labelers based on their performance, utilizing an intermediate reliability map. This map enables the model to prioritize information from reliable labelers while disregarding noisy labels. Consequently, this methodology diverges from conventional CNN-based segmentation models by eliminating the need for direct supervision of labeler performance, thereby enhancing its generalizability and adaptability across diverse datasets and labeler cohorts.

The primary contributions of this research to the field of ISS in medical images are as follows:

- The development of a novel loss function for ISS tasks. This function can infer the best possible segmentation and codify labeler reliability without needing separate inputs regarding labeler performance or the training of multiple networks.
- The creation of a Chained Deep Learning model for ISS in medical images. This model effectively learns from inconsistent labels and demonstrates improved segmentation performance, outperforming state-of-the-art approaches on a Triple Negative Breast Cancer histopathology dataset.
- A mechanism for emulating noisy annotations. This involves using network encoding layer weight disturbances to construct a crowdsourced version of the well-known Oxford-IIIT Pet Dataset.

- 761 • A Python package facilitating the use of the proposed loss function and  
762 architecture within CNN models for ISS tasks in medical images.<sup>11</sup>
- 763 • Mappings of the Triple Negative Breast Cancer and Oxford-IIIT Pet Datasets as  
764 lazy loaders, integrated into the package for immediate use with the proposed  
765 loss function.<sup>12</sup>
- 766 • A publicly accessible Github repository containing the source code developed  
767 for this work.<sup>13</sup>

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<sup>11</sup>[https://pypi.org/project/seg\\_tgce/](https://pypi.org/project/seg_tgce/)

<sup>12</sup><https://seg-tgce.readthedocs.io/en/latest/experiments.html>

<sup>13</sup>[https://github.com/blotero/seg\\_tgce](https://github.com/blotero/seg_tgce)





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CHAPTER

770

TWO

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772

## CONCEPTUAL PRELIMINARIES

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### 2.1 Modern Concept of Digital Image

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A digital image is a numerical representation of a visual scene, captured through various imaging devices and stored in a computer. From a mathematical perspective, a digital image can be represented as a function  $f(x, y)$  that maps spatial coordinates  $(x, y)$  to intensity values. In the discrete domain, this function is sampled at regular intervals, creating a matrix of values known as pixels (picture elements).

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#### 2.1.1 Types of Digital Images

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##### Grayscale Images

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Grayscale images are the simplest form of digital images, where each pixel represents a single intensity value. Mathematically, a grayscale image can be represented as a 2D matrix  $I$  of size  $M \times N$ , where each element  $I(i, j)$  represents the intensity at position  $(i, j)$ . The intensity values typically range from 0 (black) to 255 (white) in 8-bit images, or from 0 to 65535 in 16-bit images.

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## 786 **Color images**

787 Color images extend the grayscale concept by representing each pixel with multiple  
788 channels, typically Red, Green, and Blue (RGB). A color image can be represented  
789 as a 3D matrix  $I$  of size  $M \times N \times 3$ , where  $I(i, j, k)$  represents the intensity of the  
790  $k$ -th color channel at position  $(i, j)$ . Other color spaces like HSV (Hue, Saturation,  
791 Value) or CMYK (Cyan, Magenta, Yellow, Key) are also commonly used in different  
792 applications.

## 793 **Multispectral images**

794 Multispectral images capture information across multiple wavelength bands  
795 beyond the visible spectrum. These images can be represented as a 3D matrix  $I$  of  
796 size  $M \times N \times B$ , where  $B$  is the number of spectral bands. Each band  $I(i, j, b)$   
797 represents the intensity at position  $(i, j)$  for the  $b$ -th spectral band. This  
798 representation is particularly useful in medical imaging, remote sensing, and  
799 scientific applications.

## 800 **3D images and volumetric data**

801 Three-dimensional images extend the concept of pixels to voxels (volume elements).  
802 A 3D image can be represented as a 3D matrix  $V$  of size  $M \times N \times D$ , where  $D$   
803 represents the depth dimension. Each voxel  $V(i, j, k)$  represents the intensity at  
804 position  $(i, j, k)$  in the 3D space. This representation is fundamental in medical  
805 imaging (CT, MRI), scientific visualization, and computer graphics.

## 806 **2.1.2 Mathematical Representations**

807 The mathematical foundation of digital images relies on several key concepts:

- 808 • **Sampling:** The process of converting a continuous image into a discrete  
809 representation. According to the Nyquist-Shannon sampling theorem, the  
810 sampling frequency must be at least twice the highest frequency present in  
811 the image to avoid aliasing.
- 812 • **Quantization:** The process of converting continuous intensity values into  
813 discrete levels. The number of quantization levels determines the image's  
814 bit depth and affects its quality and storage requirements.
- 815 • **Resolution:** The number of pixels per unit length in an image, typically  
816 measured in pixels per inch (PPI) or dots per inch (DPI).
- 817 • **Dynamic range:** The ratio between the maximum and minimum measurable  
818 light intensities in an image, often expressed in decibels (dB).

819 The mathematical representation of a digital image can be expressed as:

$$I(x, y) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \cdot \delta(x - i, y - j) \quad (2-1)$$

820 where  $I(x, y)$  is the digital image,  $f(i, j)$  represents the intensity values, and  $\delta(x -$   
821  $i, y - j)$  is the Kronecker delta function.

822 For color images, the representation extends to:

$$I(x, y) = \begin{bmatrix} I_R(x, y) \\ I_G(x, y) \\ I_B(x, y) \end{bmatrix} \quad (2-2)$$

823 where  $I_R$ ,  $I_G$ , and  $I_B$  represent the red, green, and blue channels respectively.

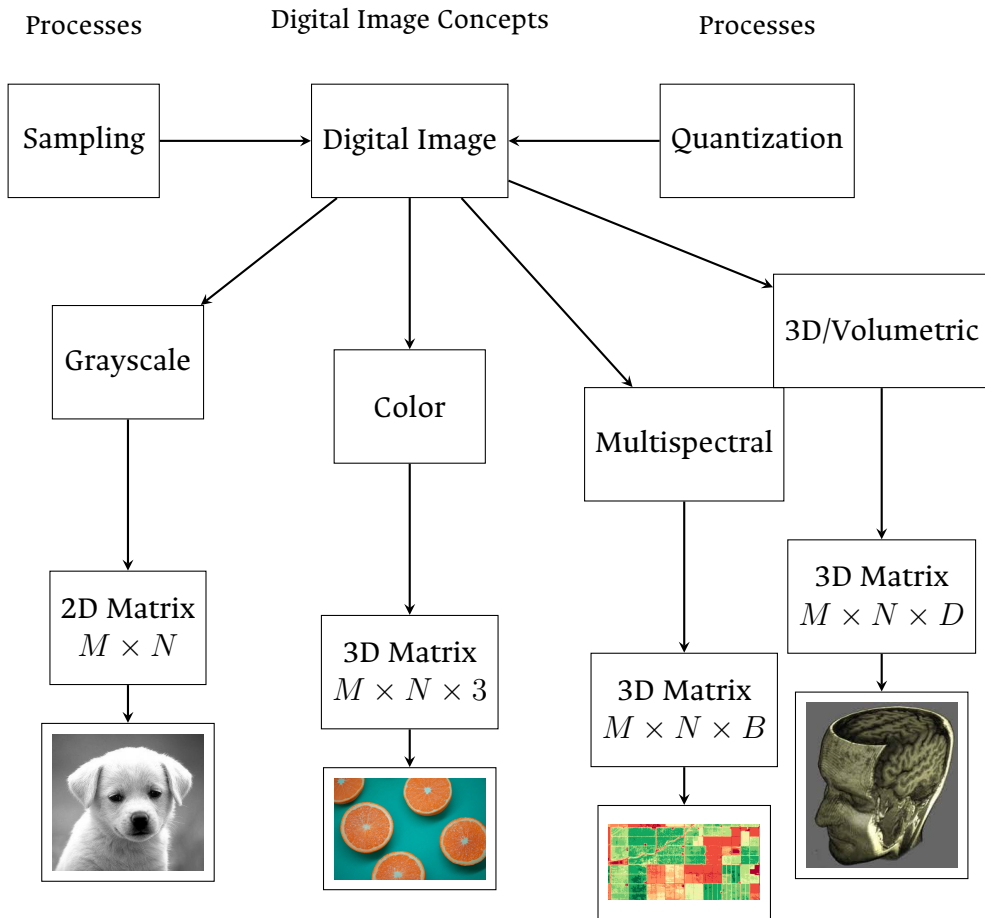


Figure 2-1 Overview of digital image concepts and their mathematical representations. The figure shows the main types of digital images (grayscale, color, multispectral, and volumetric), their mathematical representations, and the fundamental processes of sampling and quantization. Example images are included to illustrate each type.

## 2.2 Digital Histopathological Images

Digital histopathology represents a significant advancement in medical imaging, where traditional glass slides containing tissue samples are digitized using specialized scanning devices. This transformation has revolutionized the field of pathology by enabling remote diagnosis, computer-aided analysis, and digital archiving of tissue samples [Amgad et al., 2019]. This process has evolved significantly over the past few decades, as shown in Figure 2-2.

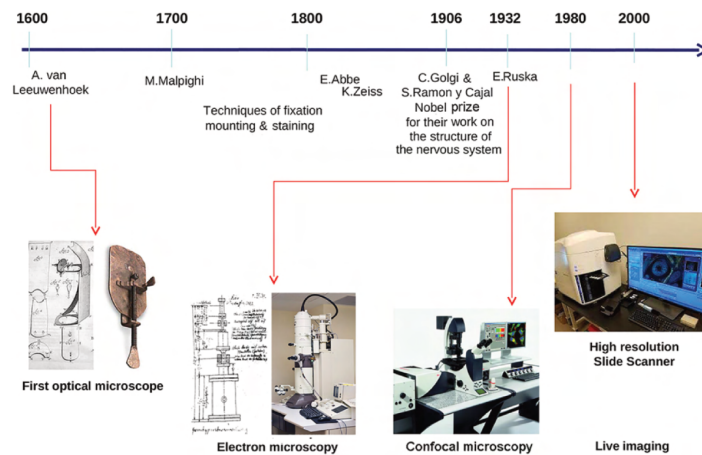


Figure 2-2 Histology evolution timeline. (Image from [Mazzarini et al., 2021]).

### 2.2.1 Whole Slide Imaging (WSI)

Whole Slide Imaging (WSI) is the process of digitizing entire glass slides at high resolution, creating a digital representation that can be viewed, analyzed, and shared electronically. Modern WSI scanners use sophisticated optical systems that capture multiple fields of view at high magnification, which are then stitched together to create a seamless digital image [Hu et al., 2025]. These systems incorporate high-resolution objectives with magnifications ranging from 20x to 40x, precise motorized stages for accurate slide positioning, automated focus

839 systems to maintain image quality, and high-quality cameras equipped with large  
840 sensor arrays. The resulting digital slides can reach sizes of several gigabytes,  
841 containing billions of pixels that capture the microscopic details of tissue samples  
842 [Hu et al., 2025]. Figure 2-3 shows a whole slide imaging system by Omnyx for  
843 slide digitization and a comprehensive digital pathology interface from Omnyx  
844 designed to streamline pathologists' diagnostic workflow [Farahani et al., 2015].

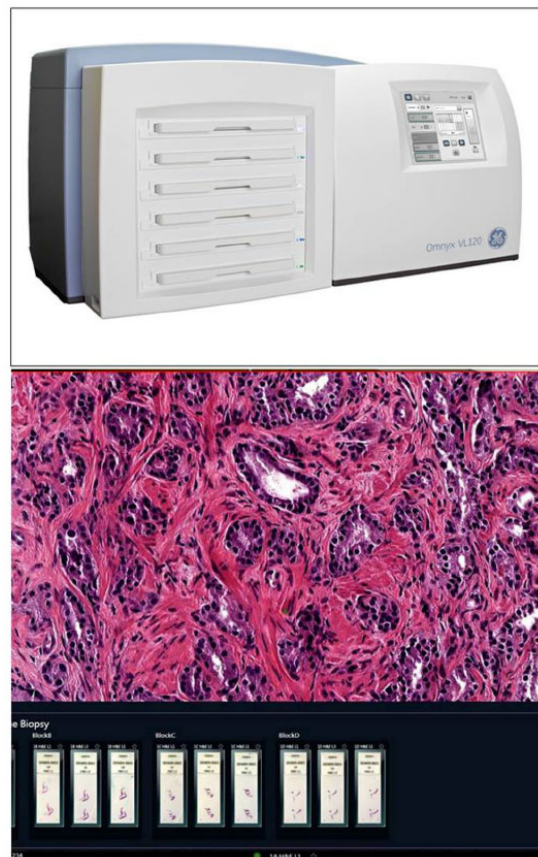


Figure 2-3 (Above) Whole slide imaging system by Omnyx for slide digitization. (Below) Comprehensive digital pathology interface from Omnyx designed to streamline pathologists' diagnostic workflow. (From [Farahani et al., 2015]).

### 845 2.2.2 Regions of Interest (ROI)

846 In digital histopathology, **Region of Interests (ROIs)** are specific areas within a  
847 whole slide image that contain diagnostically relevant information. These regions  
848 can be manually annotated by pathologists, automatically detected using  
849 computer vision algorithms, or defined based on specific tissue characteristics or  
850 abnormalities. The importance of **ROIs** lies in their ability to focus computational  
851 analysis on relevant areas, reduce computational complexity in automated  
852 systems, facilitate targeted diagnosis and research, and enable efficient storage and  
853 transmission of critical information.

### 854 2.2.3 Staining Techniques

855 Histopathological analysis relies heavily on various staining techniques to enhance  
856 the visibility of different tissue components and cellular structures. The choice of  
857 staining method depends on the specific diagnostic requirements and the type of  
858 tissue being examined.

#### 859 Hematoxylin and Eosin (H&E)

860 Hematoxylin and Eosin (H&E) staining is the most widely used technique in  
861 histopathology, particularly in breast cancer diagnosis [Pan et al., 2021]. This  
862 staining method provides essential visualization through two components:  
863 hematoxylin, which stains cell nuclei blue/purple to highlight nuclear morphology,  
864 and eosin, which stains cytoplasm and extracellular matrix pink/red to reveal  
865 tissue architecture.

866 The popularity of H&E staining in breast cancer histopathology stems from its  
867 ability to clearly visualize tumor architecture and growth patterns, distinguish  
868 between different types of breast cancer, identify important diagnostic features

like nuclear pleomorphism, and assess tumor grade and stage. Beyond breast cancer, H&E staining finds extensive application across various medical specialties including general pathology, dermatology, gastroenterology, neurology, and oncology.

## Special Stains

In addition to H&E, various special stains are used for specific diagnostic purposes. Immunohistochemistry (IHC) uses antibodies to detect specific proteins, playing a crucial role in subtyping breast cancers. Key IHC stains include Estrogen Receptor (ER) staining for detecting estrogen receptors, Progesterone Receptor (PGR) staining for assessing progesterone receptor status, Human Epidermal Growth Factor Receptor 2 (HER2) staining for evaluating HER2 protein expression, and Ki67 staining for measuring cellular proliferation rates. These markers are particularly crucial in breast cancer diagnosis and treatment planning, as they help determine the molecular subtype of the cancer and guide personalized therapeutic approaches. Other specialized stains include Periodic Acid-Schiff (PAS) for highlighting carbohydrates and basement membranes, Masson's Trichrome for distinguishing between collagen and muscle fibers, and silver stains for detecting microorganisms and nerve fibers. These specialized staining techniques complement H&E by providing additional diagnostic information that is crucial for accurate diagnosis and treatment planning [Weitz et al., 2023]. Examples of these staining techniques are shown in Figure 2-7.

## 2.3 Deep Learning Fundamentals

Deep Learning has emerged as a powerful subset of machine learning, revolutionizing the field of artificial intelligence. Its roots can be traced back to the early development of artificial neural networks in the 1940s and 1950s, with significant milestones including the perceptron in 1958 and the backpropagation



algorithm in the 1980s. However, it wasn't until the early 21st century, with the advent of more powerful computational resources and the availability of large datasets, that deep learning truly began to flourish.

### 2.3.1 Learning Paradigms

Deep Learning systems can be categorized into three main learning paradigms. The most common approach is supervised learning, where models learn from labeled data by mapping inputs to known outputs. This paradigm requires a large amount of labeled training data, which can be expensive and time-consuming to acquire. Semi-supervised learning offers a hybrid approach that leverages both labeled and unlabeled data, proving particularly useful when labeled data is scarce but unlabeled data is abundant. Finally, unsupervised learning enables models to discover patterns and structures from unlabeled data without explicit guidance, making it valuable for tasks like clustering and dimensionality reduction.

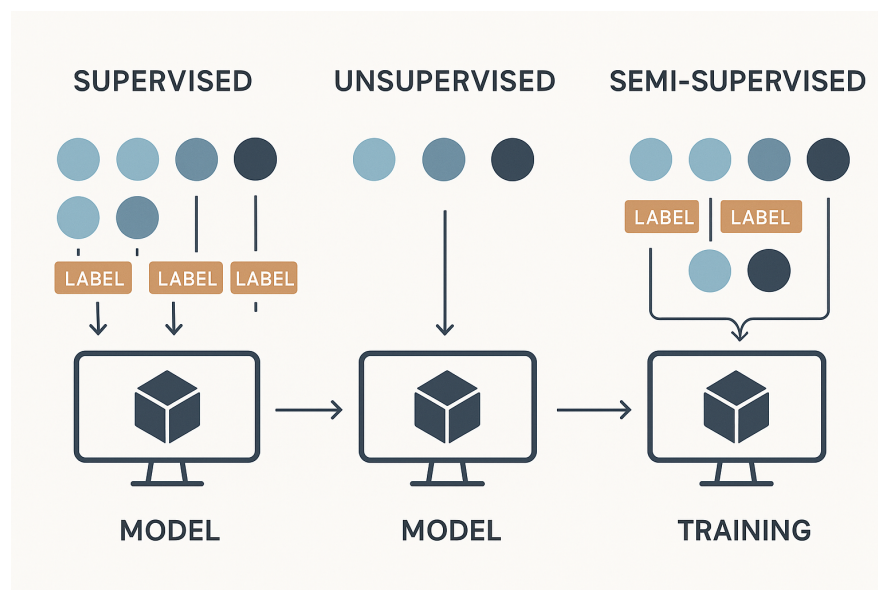


Figure 2-4 Common learning paradigms.

## 908 2.3.2 Architecture and Training

909 Deep Learning architectures are characterized by their layered structure, where  
910 each layer progressively extracts and transforms features from the input data. The  
911 early layers typically focus on low-level feature extraction, such as edges, textures,  
912 and basic patterns in the case of image processing. As information flows through  
913 the network, middle layers combine these basic features into more complex  
914 representations. The final layers perform high-level reasoning and make the  
915 ultimate predictions or classifications.

916 The training process relies heavily on the gradient descent algorithm, which  
917 iteratively adjusts the model's parameters to minimize a loss function. This loss  
918 function serves as a crucial component of the learning process, quantifying how  
919 well the model's predictions match the actual targets. By providing a measure of  
920 the model's performance, the loss function guides the optimization process,  
921 enabling the network to learn meaningful patterns from the training data.

## 922 2.3.3 Challenges and Solutions

923 Despite their power, deep learning systems face several significant challenges. One  
924 of the most prominent issues is overfitting, where models may memorize training  
925 data instead of learning generalizable patterns. This challenge is typically  
926 addressed through various regularization techniques such as dropout, L1/L2  
927 regularization, and early stopping. Another critical challenge is the substantial  
928 data requirements; deep learning models often need massive amounts of training  
929 data to achieve good performance, which can be a limiting factor in many  
930 applications. Additionally, the complex, layered nature of deep learning models  
931 makes them difficult to interpret, often referred to as "black boxes." This lack of  
932 transparency can be particularly problematic in critical applications where  
933 understanding the decision-making process is essential.

### 934 2.3.4 Deep Learning Frameworks

935 The development of powerful open-source frameworks has significantly  
936 accelerated deep learning research and applications. TensorFlow, developed by  
937 Google, provides a comprehensive ecosystem for building and deploying machine  
938 learning models [Abadi et al., 2016]. PyTorch, created by Facebook’s AI Research  
939 lab, offers dynamic computation graphs and has become particularly popular in  
940 research settings. Caffe, known for its speed and modularity, is widely used in  
941 computer vision applications.

942 These frameworks have democratized deep learning by providing efficient  
943 implementations of common operations, automatic differentiation for gradient  
944 computation, and GPU acceleration for faster training. They also offer pre-trained  
945 models and transfer learning capabilities, along with active communities for  
946 support and knowledge sharing. The combination of these frameworks with  
947 modern hardware has enabled researchers and practitioners to develop  
948 increasingly sophisticated models, pushing the boundaries of what’s possible in  
949 artificial intelligence. As shown in Figure 2-5, which presents data from Google  
950 Trends over the last five years (as of April 2025), TensorFlow and PyTorch have  
951 emerged as the two most prominent frameworks in the deep learning landscape.

## 952 2.4 Datasets and Data Sources

953 Throughout the development of this work, multiple datasets were used for  
954 evaluation of ISS models. The common elements of all these datasets are that they  
955 contain RGB images and are crowdsourced with multiple labelers, where not  
956 necessarily all labeler label all images.

957 As it has been mentioned in Chapter 1, the main goal of this work is mainly  
958 focused on crowdsourced histopathology images semantic segmentation, however,  
959 these datasets present the following challenges:

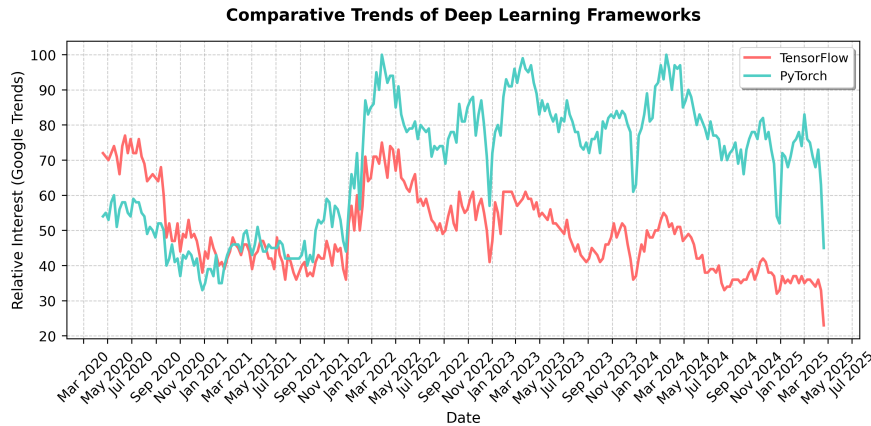


Figure 2-5 Comparative Trends of the top two most popular Deep Learning Frameworks, apparently, tendency was switched to PyTorch since 2022

- Distribution of segmentation labels is not uniform across the image, since some tissues and structures are more common than others.
- Visualization of performance of the models in debug time (like per epoch analysis) is not simple for non experts in the subject, which makes it hard to evaluate whether the model is overfitting or not at a glance.

For these reasons, multiple datasets were created in the pursuit of an initial evaluation of performance of the models against more traditional and familiar images before the focus on histopathology images. Once a decent performance in metrics like Dice coefficient was achieved, the focus was shifted to histopathology images and further tunings on the models were performed if needed.

In any case, both the emulated noisy annotations datasets and the histopathology datasets somehow contained ground truth aggregation, either from the original source (in the case of emulated noisy annotations), the expert annotation (if available) or from the aggregation of multiple labelers <sup>1</sup>.

<sup>1</sup>STAPLE in the case of histopathology datasets with no expert annotations available

### 2.4.1 Oxford-IIIT Pet Dataset

Using the techniques described later in Section 5.3.1, the Oxford-IIIT Pet Dataset [Parkhi et al., 2012] was used to create a dataset with emulated noisy annotations. This dataset is officially available in the ‘tensorflow\_datasets’ package, making it convenient to use in experiments. It is composed of 7349 images of pets, being 2371 of these images cats, and the rest 4978, dogs. Breed labels are included as well as segmentation masks for the classes: ‘background’, ‘border’, and ‘pet’. Figure 2-6 shows an example of the annotations in the original dataset.

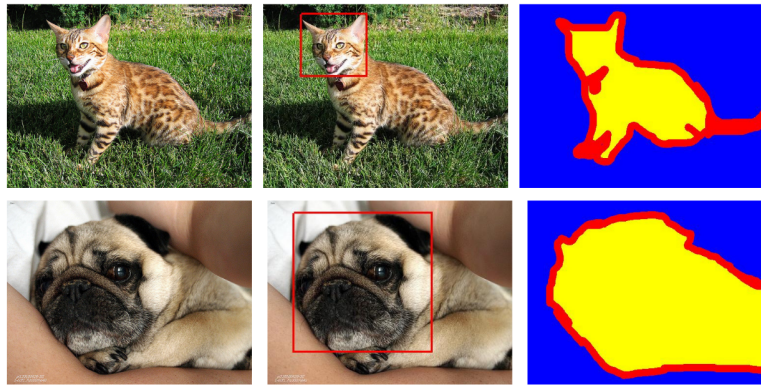


Figure 2-6 Annotations in the Oxford-IIIT Pet data. From left to right: pet image, head bounding box, and trimap segmentation (blue: background region; red: ambiguous region; yellow: foreground region).

### 2.4.2 Real histopathology datasets

#### Multi-Stain Breast Cancer Histological Dataset

The Multi-Stain Breast Cancer Histological Dataset [Weitz et al., 2023] represents one of the largest publicly available collections of whole slide images (WSIs) from surgical resection specimens of primary breast cancer patients. This dataset is particularly valuable for our work because it contains matched pairs of H&E and

IHC-stained tissue sections from the same tumor, with a total of 4,212 WSIs from 1,153 patients. The IHC stains include important biomarkers such as ER, PGR, HER2, and KI67, which are routinely used in breast cancer diagnosis and treatment planning (more on staining techniques in Section 2.2.3).

The dataset’s relevance to our work stems from several key aspects. The matched H&E and IHC stains allow for studying the consistency of segmentation across different staining modalities, which is crucial for understanding how different visualization methods affect annotation quality. With 1,153 patients, the dataset provides a robust foundation for training and evaluating segmentation models in a real-world clinical setting. The inclusion of routine diagnostic cases makes the dataset representative of actual clinical practice, where variations in staining quality and tissue preparation are common. Furthermore, the multiple biomarker stains (ER, PGR, HER2, KI67) enable the study of how different tissue characteristics affect segmentation performance and annotator agreement.

This dataset serves as an ideal testbed for our crowdsourced segmentation approach. It allows us to evaluate how different staining modalities affect annotator performance and agreement, while also providing insights into the relationship between tissue characteristics and segmentation difficulty. The dataset’s comprehensive nature enables validation of our models’ performance across different biomarker expressions and assessment of the generalizability of our approach to real-world clinical data.

## Structured Crowdsourcing Dataset for Histology Images

The dataset presented by [Amgad et al., 2019] is particularly relevant to our work as it represents one of the first systematic studies of crowdsourced annotations in histopathology. The authors recruited 25 participants with varying levels of expertise (from senior pathologists to medical students) to delineate tissue regions in 151 breast cancer slides using the Digital Slide Archive platform, resulting in over 20,000 annotated tissue regions.

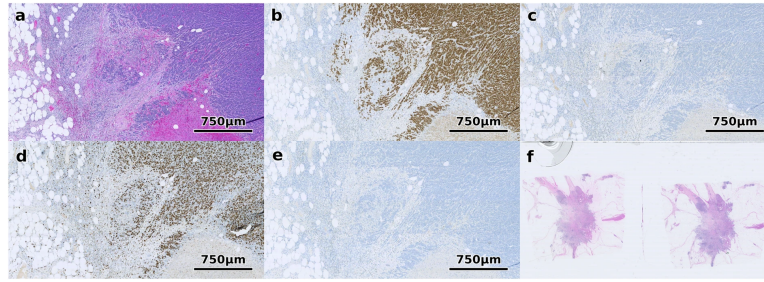


Figure 2-7 Different staining techniques obtained from multi-stain breast cancer dataset [Weitz et al., 2023]. (a) shows H&E, (b) ER, (c) HER2, (d) Ki67 and (e) PGR. (f) shows an example of a WSI that was excluded since it contains multiple tissue sections.

Key aspects of this dataset make it valuable for our work. The systematic evaluation of inter-participant discordance revealed varying levels of agreement across different tissue classes, with low discordance for tumor and stroma, and higher discordance for more subjectively defined or rare tissue classes. The inclusion of feedback from senior participants helped in curating high-quality annotations, demonstrating that fully convolutional networks trained on these crowdsourced annotations can achieve high accuracy (mean AUC=0.945). The dataset also provides evidence that the scale of annotation data significantly improves image classification accuracy.

This dataset is particularly valuable for our work because it provides crucial insights into how annotator expertise affects segmentation quality. It demonstrates the feasibility of using crowdsourced annotations for training accurate segmentation models, showing that even with varying levels of expertise, aggregated annotations can produce reliable ground truth. The dataset includes a systematic analysis of inter-annotator agreement, which is crucial for understanding the challenges in crowdsourced histopathology segmentation and informing the development of more robust segmentation approaches.



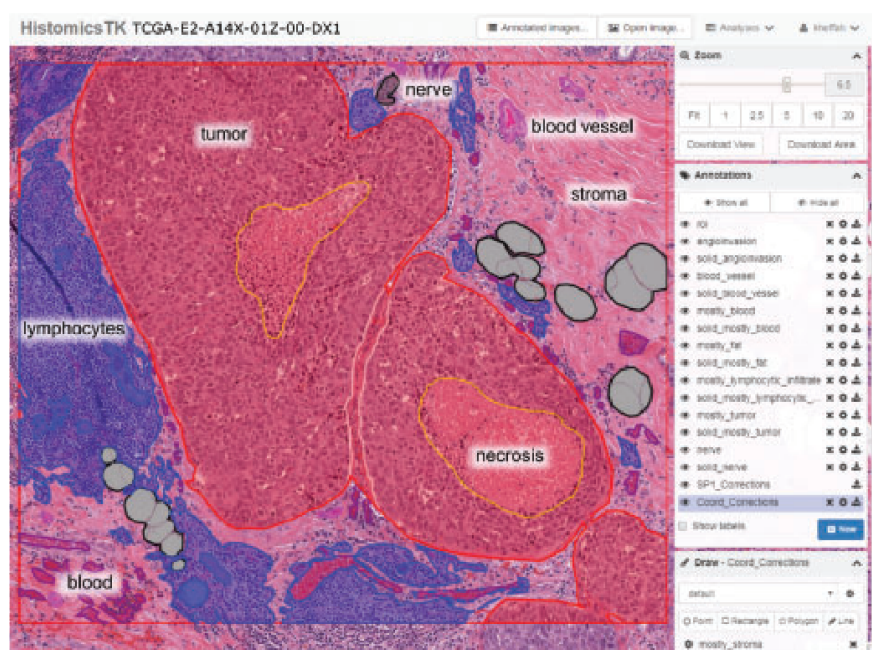


Figure 2-8 Screenshot of the DSA and HistomicsTK web interface while creating the crowdsourced annotations for the dataset presented by [Amgad et al., 2019].



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CHAPTER

**THREE**

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## CHAINED GAUSSIAN PROCESSES

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### 3.1 Background and Related Methods

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In recent years, several approaches have been proposed to handle supervised learning problems in the context of multiple annotators. This section reviews key methods that form the foundation for understanding chained Gaussian processes and their application to multiple annotator scenarios in regression and classification tasks.

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#### 3.1.1 Kernel Alignment-Based Annotator Relevance Analysis (KAAR)

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The Kernel Alignment-Based Annotator Relevance Analysis (KAAR) method, introduced in [Julián and Álvarez Meza Andrés Marino, 2023], addresses the challenge of estimating annotator expertise in scenarios where the ground truth is

1049 unavailable. The key innovation of KAAR lies in its use of Centered Kernel  
 1050 Alignment (CKA) to measure the similarity between input features and annotator  
 1051 labels.

1052 Given a dataset with input features  $\mathbf{X}$  and labels from multiple annotators  $\mathbf{Y}$ , KAAR  
 1053 computes a kernel matrix  $\mathbf{K}_\nu$  as a convex combination of  $R$  basis kernels:

$$\mathbf{K}_\nu = \sum_{r=1}^R \nu_r \mathbf{K}_r, \quad (3-1)$$

1054 where  $\mathbf{K}_r$  is computed using a kernel function over the input features, and  $\nu_r$   
 1055 represents the relevance weight for the  $r$ -th annotator. The weights are optimized  
 1056 by maximizing the CKA between  $\mathbf{K}_\nu$  and a target kernel  $\mathbf{F}$  computed from the  
 1057 input features:

$$\rho(\mathbf{K}_\nu, \mathbf{F}) = \frac{\langle \bar{\mathbf{K}}_\nu, \bar{\mathbf{F}} \rangle_{\text{F}}}{\|\bar{\mathbf{K}}_\nu\|_{\text{F}} \|\bar{\mathbf{F}}\|_{\text{F}}}, \quad (3-2)$$

1058 A key advantage of KAAR is its ability to capture dependencies between annotators  
 1059 through the kernel framework, without requiring explicit parametric modeling of  
 1060 these relationships.

### 1061 3.1.2 Localized Kernel Alignment-Based Annotator 1062 Relevance Analysis (LKAAR)

1063 LKAAR extends KAAR by introducing locality in the kernel alignment framework.  
 1064 Rather than assuming constant annotator performance across the input space,

1065 LKAAR models the annotator relevance as a function of the input features. The  
 1066 kernel combination in LKAAR takes the form:

$$K_q = \sum_{r=1}^R Q_r K_r Q_r, \quad (3-3)$$

1067 where  $Q_r$  is a diagonal matrix whose elements represent the local relevance of  
 1068 annotator  $r$ . These elements are computed as:

$$q_r(\mathbf{x}_n) = \begin{cases} \beta_0^{(r)} + \sum_{n'=1}^N \beta_{n'}^{(r)} \kappa_\beta(\mathbf{x}_n, \mathbf{x}_{n'}), & \text{if } n \in N_r \\ 0, & \text{Otherwise} \end{cases} \quad (3-4)$$

1069 This localized approach allows LKAAR to better model inconsistent annotators  
 1070 whose performance varies across different regions of the input space.

### 1071 3.1.3 Regularized Chained Deep Neural Network 1072 (RCDNN)

1073 RCDNN represents a deep learning approach to multiple annotator learning,  
 1074 inspired by the chained Gaussian processes model. The method models both the  
 1075 ground truth estimation and annotator reliability through a deep neural network  
 1076 architecture. The likelihood function in RCDNN takes the form:

$$p(\mathbf{Y}|\boldsymbol{\theta}) = \prod_{n=1}^N \prod_{r \in R_n} \left( \prod_{k=1}^K \zeta_{n,k}^{\delta(y_n^{(r)} - k)} \right)^{\lambda_n^{(r)}} \left( \frac{1}{K} \right)^{1 - \lambda_n^{(r)}}, \quad (3-5)$$

1077 where  $\lambda_n^{(r)}$  represents the reliability of annotator  $r$  for instance  $n$ , and  $\zeta_{n,k}$  is the  
 1078 estimated probability of class  $k$  for instance  $n$ .

1079 The RCDNN architecture is built upon a sophisticated multi-layer structure that  
 1080 combines several key innovations in deep learning. At its core, it employs multiple  
 1081 dense layers with carefully chosen activation functions, each designed to capture  
 1082 different aspects of the annotator-data relationship. To prevent overfitting, a  
 1083 comprehensive regularization strategy is implemented, combining both l1 and l2  
 1084 norms with dropout mechanisms. The network architecture splits into specialized  
 1085 branches: one dedicated to estimating the ground truth labels, and another  
 1086 focused on modeling annotator reliability. This dual-branch approach allows the  
 1087 network to simultaneously learn both the true underlying patterns in the data and  
 1088 the individual characteristics of each annotator. Furthermore, the implementation  
 1089 of Monte Carlo dropout during prediction provides a mechanism for uncertainty  
 1090 estimation, offering not just predictions but also confidence measures for those  
 1091 predictions.

### 1092 3.1.4 Chained Gaussian Processes Model

1093 The Chained Gaussian Processes (CGP) model provides a probabilistic framework for  
 1094 modeling multiple outputs through a chain of Gaussian processes. In the context of  
 1095 multiple annotators, CGP can be used to model both the ground truth and annotator  
 1096 behavior. The key idea is to model a likelihood function with  $J$  parameters as in  
 1097 [Giraldo et al., 2022]:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{n=1}^N p(y_n|\theta_1(\mathbf{x}_n), \dots, \theta_J(\mathbf{x}_n)), \quad (3-6)$$

1098 where each parameter  $\theta_j(\mathbf{x})$  is modeled using a separate Gaussian process. This  
 1099 framework provides remarkable flexibility in modeling complex relationships

1100 within the data. The use of Gaussian processes enables natural uncertainty  
1101 quantification through posterior distributions, allowing the model to express  
1102 varying degrees of confidence in its predictions. The framework inherently  
1103 handles missing data through its probabilistic nature, and the automatic relevance  
1104 determination property of GPs helps identify the most important features for  
1105 prediction.

1106 The CGP framework becomes particularly powerful when integrated with concepts  
1107 from KAAR and LKAAR. By incorporating kernel alignment principles, the model  
1108 can make more informed choices about kernel functions, better capturing the  
1109 underlying structure of the data. The framework can be extended to include  
1110 input-dependent lengthscales, allowing for varying degrees of smoothness across  
1111 the input space. Furthermore, through multi-output GP formulations, the model  
1112 can capture complex dependencies between different annotators, providing a rich  
1113 representation of how different experts' opinions relate to each other.

1114 However, implementing CGP in practice presents several significant challenges.  
1115 The computational complexity of the model scales cubically with the dataset size,  
1116 making it potentially prohibitive for large-scale applications. The need for  
1117 approximate inference techniques in non-Gaussian likelihood scenarios can  
1118 introduce additional complexity and potential approximation errors. When  
1119 dealing with high-dimensional inputs, the model may struggle with the curse of  
1120 dimensionality, requiring careful feature selection or dimensionality reduction  
1121 preprocessing. Additionally, the selection of appropriate kernel functions remains  
1122 a critical challenge, as it significantly impacts the model's performance and its  
1123 ability to capture relevant patterns in the data.

## 1124 3.2 Tooling and Implementation

1125 The implementation of Chained Gaussian Processes for multiple annotator  
1126 learning requires robust and flexible software tools that can handle both the

1127 probabilistic nature of Gaussian processes and the complexity of deep  
1128 architectures. In our implementation, we leverage two key frameworks: GPflow  
1129 and GPflux. These tools provide the foundation for building scalable and efficient  
1130 implementations of our models while maintaining the theoretical rigor necessary  
1131 for proper uncertainty quantification.

1132 The choice of these frameworks is motivated by several factors. First, both are  
1133 built on top of TensorFlow, providing access to automatic differentiation and GPU  
1134 acceleration capabilities. Second, they offer a modular design that allows for easy  
1135 extension and modification, which is crucial when implementing novel model  
1136 architectures. Third, they provide robust implementations of various Gaussian  
1137 process models and inference methods, allowing us to focus on the specific  
1138 challenges of multiple annotator learning rather than reimplementing basic GP  
1139 functionality.

### 1140 3.2.1 GPflow Overview

1141 GPflow is a Python package for building Gaussian process models in TensorFlow.  
1142 Developed as a modern and scalable alternative to GPy, it leverages TensorFlow’s  
1143 computational capabilities to provide efficient implementation of Gaussian process  
1144 models. The framework is designed with four key principles in mind: ease of use,  
1145 computational efficiency, extensibility, and automatic differentiation. This work  
1146 was originally published in [de G. Matthews et al., 2016].

1147 At its core, GPflow provides implementations of various Gaussian process models  
1148 including GPR (Gaussian Process Regression), GPMC (Gaussian Process Monte  
1149 Carlo), SGPR (Sparse Gaussian Process Regression), and SGPMC (Sparse Gaussian  
1150 Process Monte Carlo). These implementations are built on top of TensorFlow’s  
1151 computational graph framework, allowing for automatic differentiation and GPU  
1152 acceleration where available.

1153 The framework’s architecture is modular, with clear separation between model  
1154 components such as kernels, likelihoods, and optimization routines. This  
1155 modularity makes it straightforward to extend existing models or create new ones.  
1156 GPflow includes a comprehensive suite of kernel functions, ranging from the basic  
1157 RBF (Radial Basis Function) kernel to more sophisticated ones like the Spectral  
1158 Mixture kernel.

1159 One of GPflow’s distinguishing features is its robust handling of model parameters.  
1160 It provides facilities for parameter transformation (ensuring parameters remain in  
1161 valid ranges during optimization), prior specification, and flexible optimization  
1162 strategies. The framework also includes utilities for model evaluation and  
1163 prediction, with built-in support for uncertainty estimation.

## 1164 3.2.2 GPflux Framework

1165 GPflux extends GPflow to enable the construction and training of Deep Gaussian  
1166 Process (DGP) models using modern deep learning practices. Built on top of both  
1167 GPflow and TensorFlow, GPflux provides a flexible framework for implementing  
1168 deep probabilistic models that combine the expressiveness of deep neural  
1169 networks with the uncertainty quantification capabilities of Gaussian processes.  
1170 This work was originally published in [Dutordoir et al., 2021].

1171 The framework implements Deep Gaussian Processes as a composition of Gaussian  
1172 process layers, where each layer transforms its inputs through a Gaussian process  
1173 mapping. This hierarchical structure allows the model to capture complex,  
1174 non-stationary patterns in data while maintaining uncertainty estimates  
1175 throughout the entire processing pipeline. GPflux handles this complexity through  
1176 a carefully designed architecture that separates the concerns of model  
1177 specification, inference, and prediction.

1178 A key innovation in GPflux is its integration with Keras-style layers, allowing  
1179 practitioners to combine deterministic neural network layers with Gaussian

process layers in a single model. This hybrid approach enables the construction of models that can leverage both the structural assumptions encoded in neural network architectures and the flexibility of Gaussian processes.

For inference, GPflux implements various approximate methods, including stochastic variational inference with inducing points, which makes it possible to train deep Gaussian process models on large datasets. The framework provides built-in support for minibatch training, making it practical to work with datasets that don't fit in memory.

## 3.3 Experimental Setup

Synthetic datasets were used to evaluate the proposed methods in classification and regression tasks, following very similar procedures to the ones used in [Gil-Gonzalez et al., 2023], which are as follows:

### 3.3.1 Classification

A fully synthetic data for a 1-D ( $P = 1$ ) multiclass classification problem with  $K = 3$  classes is generated. The input feature matrix  $\mathbf{X}$  is constructed by randomly sampling  $N = 100$  points from a uniform distribution over the interval  $[0, 1]$ . For each sample  $n$ , the true label is determined by computing three target values:

$$\begin{aligned} t_{n,1} &= \sin(2\pi x_n), \\ t_{n,2} &= -\sin(2\pi x_n), \\ t_{n,3} &= -\sin(2\pi(x_n + 0.25)) + 0.5, \end{aligned} \tag{3-7}$$



and assigning the class label as the index  $i \in \{1, 2, 3\}$  that maximizes  $t_{n,i}$ . This construction ensures a nontrivial, nonlinear decision boundary between the classes.

For evaluation, the test set is generated by extracting 200 equally spaced samples from the interval  $[0, 1]$ , providing a dense coverage of the input space to assess generalization performance.

Note that at this point, the fully synthetic dataset does not contain real annotator labels. Therefore, it is necessary to simulate annotator labels as corrupted versions of the hidden ground truth. In these experiments, the simulation is designed to account for: 1) dependencies among annotators, and 2) labeler performance that varies as a function of the input features.

To achieve this, an **SLFM**-based approach (termed SLFM-C) is employed, which generates annotator labels as follows:

1. **Definition of deterministic functions:** Define  $Q$  deterministic functions  $\hat{\mu}_q : \mathcal{X} \rightarrow \mathbb{R}$  and their combination parameters  $\hat{w}_{l,r,q} \in \mathbb{R}$ , for all  $r \in R$ ,  $n \in N$ .
2. **Annotator performance computation:** Compute  $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$ , where  $\hat{x}_n \in \mathbb{R}$  is the  $n$ th component of  $\hat{\mathbf{x}} \in \mathbb{R}^N$ . Here,  $\hat{\mathbf{x}}$  is the 1-D representation of the input features in  $\mathbf{X}$ , obtained using the t-distributed Stochastic Neighbor Embedding (t-SNE) approach.
3. **Reliability calculation:** Calculate  $\hat{\lambda}_n^r = \varsigma(\hat{f}_{l,r,n})$ , where  $\varsigma(\cdot) \in [0, 1]$  is the sigmoid function.
4. **Label assignment:** The  $r$ th annotator's label for the  $n$ th sample is given by

$$y_n^r = \begin{cases} y_n, & \text{if } \hat{\lambda}_n^r \geq 0.5 \\ \tilde{y}_n, & \text{if } \hat{\lambda}_n^r < 0.5 \end{cases}$$

where  $y_n$  is the true label and  $\tilde{y}_n$  is a flipped version of the actual label  $y_n$ .

This procedure ensures that the simulated annotator labels exhibit both inter-annotator dependencies and input-dependent reliability.

### 1221 3.3.2 Regression

1222 A fully synthetic dataset is also generated for a 1-D regression problem, where the  
1223 ground truth for the  $n$ th sample is defined as:

$$y_n = \sin(2\pi x_n) \sin(6\pi x_n) \quad (3-8)$$

1224 Here, the input matrix  $\mathbf{X}$  is constructed by randomly sampling 100 points uniformly  
1225 from the interval  $[0, 1]$ . The test set is created by extracting equally spaced samples  
1226 from the same interval, ensuring a clear separation between training and testing  
1227 data.

1228 To simulate the presence of multiple annotators, the ground truth is corrupted with  
1229 annotator-specific noise. For each annotator  $r$ , the observed label for the  $n$ th sample  
1230 is given by:

$$y_n^r = y_n + \epsilon_n^r$$

1231 where  $\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$  is Gaussian noise, and  $v_n^r$  represents the error variance for  
1232 the  $r$ th annotator, which may depend on the input features. This setup allows us  
1233 to model both the bias and the reliability of each annotator, as well as potential  
1234 dependencies between annotators.

1235 To further increase realism, the error variance  $v_n^r$  can be modeled as a function of  
1236 the input features, capturing scenarios where annotator reliability varies across the  
1237 input space, thus:

$$\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$$

1238 being  $v_n^r$ , the  $r$ th annotator's error variance for the  $n$ th sample.

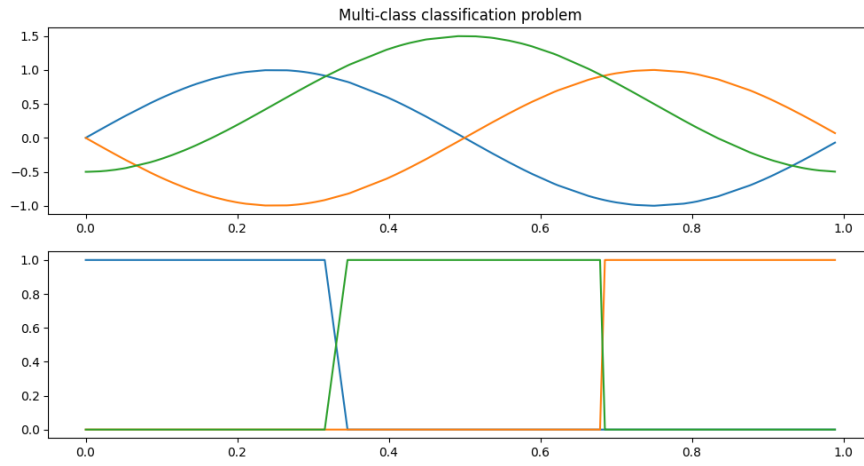
1239 Again, for modeling the error as a function of the input features, an **SLFM**-based  
1240 approach (termed SLFM-R) is employed to build the noisy labels, as follows:

- 1241 1. **Definition of deterministic functions:** Define  $Q$  deterministic functions  $\hat{\mu}_q :$   
1242  $\mathcal{X} \rightarrow \mathbb{R}$  and their combination parameters  $\hat{w}_{l,r,q} \in \mathbb{R}$ , for all  $r \in R, n \in N$ .
- 1243 2. **Annotator performance computation:** Compute  $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$ ,  
1244 where  $\hat{x}_n \in \mathbb{R}$  is the  $n$ th component of  $\hat{\mathbf{x}} \in \mathbb{R}^N$ . Here,  $\hat{\mathbf{x}}$  is the 1-D  
1245 representation of the input features in  $\mathbf{X}$ , obtained using the t-distributed  
1246 Stochastic Neighbor Embedding (t-SNE) approach.
- 1247 3. **Compute variance :** Finally, determine  $v_n^r = \exp(\hat{f}_{l,r,n})$ .

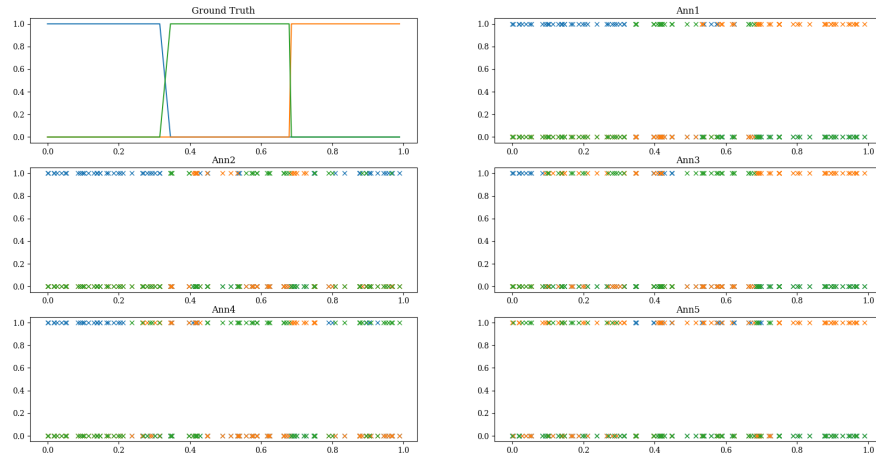
## 1248 3.4 Results

1249 The built synthetic datasets are used to train the **CCGPMA** model. Emulated  
1250 classification labels following the true label determining function from 3-7 can be  
1251 seen in Figure 3-1. On the other hand, the ground truth and noisy labels emulation  
1252 for annotators across the feature space for regression task can be seen in Figure  
1253 3-3.

1254 Using both GPflux and GPflow for the regression and classifications tasks, leads to  
1255 the following combinations experiments: GPflux-CCGPMA Regression,  
1256 GPflow-CCGPMA Regression, GPflux-CCGPMA Binary Classification,  
1257 GPflow-CCGPMA Binary Classification, GPflow-CCGPMA Multiclass Classification,  
1258 GPflux-CCGPMA Multiclass Classification. All of the experiments converged  
1259 successfully with almost identical results and close to no difference in  
1260 performance.



(a) Ground truth labels across the feature space



(b) Noisy emulations for annotators across the feature space for classification task

Figure 3-1 Synthetic dataset overview for classification task

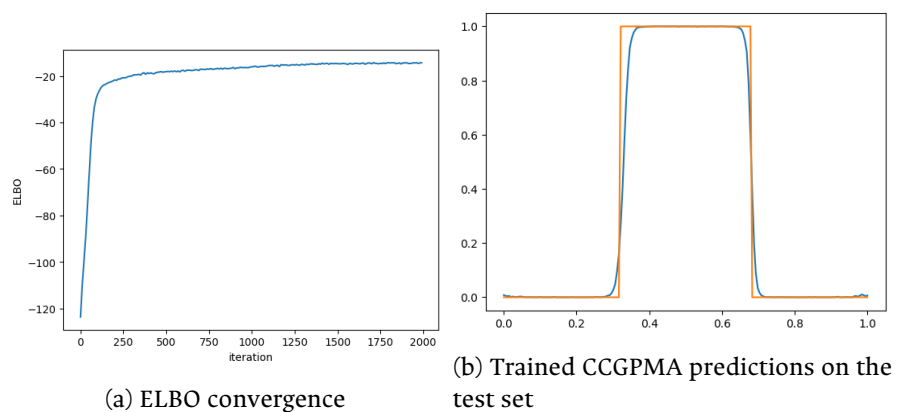


Figure 3-2 Classification results for the CCGPMA model using GPflux.

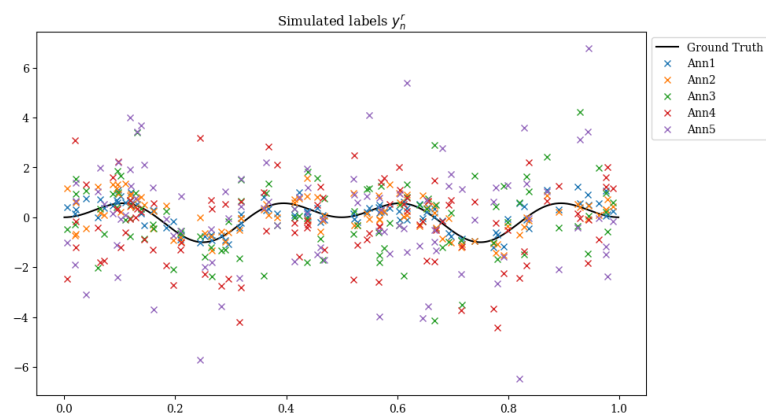
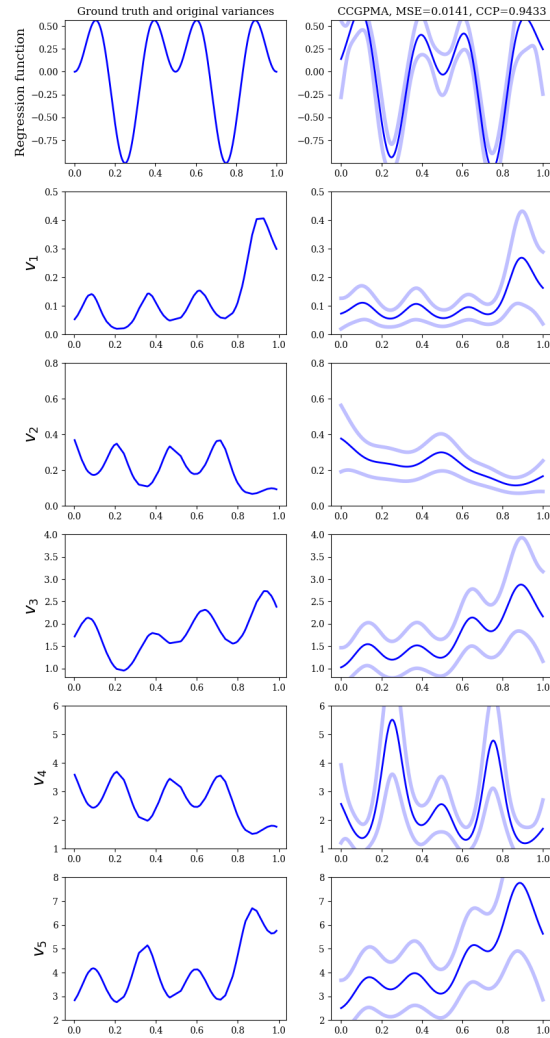


Figure 3-3 Ground truth and noisy labels emulation for annotators across the feature space for regression task.



(a) Trained CCGPMA predictions on the test set, for inference of both the hidden ground truth and the annotators' reliabilities across the feature space.

Figure 3-4 Regression results for the CCGPMA model using GPflux.

## 3.5 Conclusion

### 3.5.1 Comparative Analysis

The evolution of multiple annotator learning methods represents a fascinating progression in how we approach the challenge of learning from diverse expert opinions. KAAR established a foundation with its non-parametric approach through kernel alignment, though it operates under the assumption of constant annotator performance across the input space. LKAAR advanced this concept by introducing input-dependent annotator behavior, recognizing that expert performance often varies across different types of inputs. RCDNN brought the power and flexibility of deep learning to the problem, offering excellent scalability but requiring careful attention to regularization and architecture design. CGP represents perhaps the most theoretically elegant approach, providing a principled probabilistic framework, though it faces significant computational challenges in practical applications.

The selection of an appropriate method for a given application requires careful consideration of several factors. The size and dimensionality of the dataset play a crucial role, as some methods scale better than others with increasing data volume. The need for uncertainty quantification may favor probabilistic approaches like CGP, while computational constraints might push towards more efficient methods like KAAR. The presence of input-dependent annotator behavior could make LKAAR or RCDNN more appropriate choices. Additionally, when interpretability is a key requirement, as in many medical applications, methods like KAAR and LKAAR might be preferred over the more complex deep learning approaches. Table 3-1 provides a comprehensive comparison of the key characteristics and capabilities of each method discussed.

Table 3-1 Comparison of Multiple Annotator Learning Methods

| Characteristic                     | KAAR | LKAAR  | RCDNN   | CGP    |
|------------------------------------|------|--------|---------|--------|
| Input-dependent annotator behavior | No   | Yes    | Yes     | Yes    |
| Annotator dependencies modeling    | Yes  | Yes    | Yes     | Yes    |
| Uncertainty quantification         | No   | No     | Partial | Yes    |
| Computational scalability          | High | Medium | High    | Low    |
| Missing data handling              | Yes  | Yes    | Yes     | Yes    |
| Non-linear relationships           | Yes  | Yes    | Yes     | Yes    |
| Training complexity                | Low  | Medium | High    | High   |
| Hyperparameter sensitivity         | Low  | Medium | High    | Medium |
| Interpretability                   | High | Medium | Low     | Medium |
| Memory requirements                | Low  | Medium | High    | High   |

**Note:** Partial uncertainty quantification in RCDNN refers to the use of Monte Carlo dropout for approximate uncertainty estimation.

## 1286 3.5.2 Implementation Considerations

1287 When implementing Chained Gaussian Processes for multiple annotator learning  
 1288 using these tools, several key considerations must be addressed. First, the model  
 1289 architecture must be carefully designed to capture both the individual annotator  
 1290 characteristics and their interdependencies. This requires extending the basic GP  
 1291 layers provided by GPflux to handle multiple output streams - one for each  
 1292 annotator's reliability model.

1293 Second, the inference procedure must be adapted to handle the unique challenges  
 1294 of multiple annotator data, such as missing labels and varying levels of annotator  
 1295 expertise across different regions of the input space. The variational inference  
 1296 capabilities of GPflux provide a solid foundation for this, but careful attention  
 1297 must be paid to the design of the variational approximation to ensure it captures  
 1298 the relevant uncertainty in both the ground truth and annotator reliability  
 1299 estimates.

1300 Third, kernel selection and design presents a critical challenge in implementing the  
 1301 CGP model. The choice of kernel functions significantly impacts the model's ability



to capture the underlying structure of the data and the relationships between annotators. Several considerations must be taken into account:

- **Kernel Complexity:** The model requires multiple kernel functions to capture different aspects of the data - one for the ground truth and additional kernels for each annotator's reliability model. This multi-kernel setup increases the complexity of kernel selection and optimization.
- **Input-Dependent Behavior:** Since annotator performance often varies across the input space, the kernel functions must be flexible enough to capture this non-stationary behavior. This may require using composite kernels or kernels with input-dependent lengthscales.
- **Inter-Annotator Dependencies:** The kernel design must account for potential correlations between different annotators' behaviors. This can be achieved through careful design of the kernel structure in the SLFM framework, where the combination of basis kernels must be chosen to effectively model these dependencies.
- **Computational Efficiency:** The choice of kernel functions also impacts computational performance. While more expressive kernels may better capture complex patterns, they often come with increased computational cost. This trade-off must be carefully considered, especially when dealing with large datasets.

Finally, computational efficiency becomes a critical concern when scaling to large datasets with multiple annotators. The minibatch training capabilities of both frameworks help address this, but additional considerations such as appropriate inducing point selection and batch size tuning become important for achieving good performance in practice.

### 1327 3.5.3 Final Remarks

1328 In this chapter, we have presented a comprehensive overview of the Chained  
1329 Gaussian Processes model and its application to multiple annotator learning. We  
1330 have discussed the theoretical underpinnings of the model, its implementation  
1331 using GPflow and GPflux, and its evaluation on synthetic datasets. It was shown in  
1332 [Gil-Gonzalez et al., 2023] that CCGPMA outperforms other method in state of the  
1333 art methods in terms of accuracy and uncertainty quantification (see 1.3.1).

1334 While Gaussian Processes have demonstrated remarkable success in regression  
1335 and classification tasks, their application to image segmentation presents  
1336 significant computational challenges. The cubic computational complexity of GPs  
1337 ( $\mathcal{O}(N^3)$ ), which scales with the number of data points, becomes particularly  
1338 problematic when dealing with high-resolution images where each pixel can be  
1339 considered a data point. This limitation makes traditional GP approaches  
1340 impractical for large-scale image segmentation tasks, where the number of pixels  
1341 can easily reach into the millions. Alternative approaches, such as deep  
1342 learning-based segmentation methods, often provide more computationally  
1343 efficient solutions for these high-dimensional problems, though at the cost of  
1344 uncertainty quantification capabilities that GPs naturally provide.

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CHAPTER

**FOUR**

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## TRUNCATED GENERALIZED CROSS ENTROPY FOR SEGMENTATION

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### **4.1 Loss Functions for Multiple Annotators**

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As mentioned in Section 2.3.2, a loss function is a key element for defining the objective function of a deep learning model. The categorical cross-entropy loss is a common loss function for classification tasks. However, in the case of multiple annotators, the categorical cross-entropy loss is not able to handle the varying reliability of the annotators. In this section, we will propose a loss function that is able to handle multiple annotators' segmentation masks while accounting for their varying reliability across different regions of the image.

### 1359 4.1.1 Generalized Cross Entropy

1360 The Generalized Cross Entropy (GCE) loss function was first introduced by [Zhang  
1361 and Sabuncu, 2018] as a robust alternative to the standard cross-entropy loss,  
1362 particularly effective in handling noisy labels. Let us first consider the Cross  
1363 Entropy (CE) and Mean Absolute Error (MAE) loss functions:

$$MAE(\mathbf{y}, f(\mathbf{x})) = \|\mathbf{y} - f(\mathbf{x})\|_1 \quad (4-1)$$

$$CE(\mathbf{y}, f(\mathbf{x})) = \sum_{k=1}^K y_k \log(f_k(\mathbf{x})) \quad (4-2)$$

1364 where  $y_k \in \mathbf{y}$ ,  $f_k(\mathbf{x}) \in f(\mathbf{x})$ , and  $\|\cdot\|_1$  stands for the  $l_1$ -norm. Of note,  $\mathbf{1}^\top \mathbf{y} =$   
1365  $\mathbf{1}^\top f(\mathbf{x}) = 1$ ,  $\mathbf{1} \in \{1\}^K$  being an all-ones vector. In addition, the MAE loss can be  
1366 rewritten for softmax outputs, yielding:

$$MAE(\mathbf{y}, f(\mathbf{x})) = 2(1 - \mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x}))) \quad (4-3)$$

1367 where  $\odot$  stands for the element-wise product.

1368 The CE loss function exhibits several distinct characteristics that make it  
1369 particularly sensitive to noisy labels. It is unbounded from above and heavily  
1370 penalizes confident but wrong predictions, which can lead to instability in the  
1371 presence of label noise. In contrast, the MAE loss offers a more robust alternative  
1372 with its bounded nature and symmetric properties in softmax-based  
1373 representations. While the MAE loss is more resilient to noisy labels, it tends to

1374 train more slowly due to its equal weighting of all mistakes regardless of  
1375 confidence.

1376 The GCE loss function, as defined by the authors in [Zhang and Sabuncu, 2018],  
1377 provides a flexible framework that bridges these two approaches:

$$GCE(\mathbf{y}, f(\mathbf{x})) = 2 \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q}, \quad (4-4)$$

1378 with  $q \in (0, 1]$ . Remarkably, the limiting case for  $q \rightarrow 0$  in GCE is equivalent to the  
1379 CE expression, and when  $q = 1$ , it equals the MAE loss. In addition, the GCE holds  
1380 the following gradient with regard to  $\theta$ :

$$\frac{\partial GCE(\mathbf{y}, f(\mathbf{x}; \theta) | k)}{\partial \theta} = -f_k(\mathbf{x}; \theta)^{q-1} \nabla_\theta f_k(\mathbf{x}; \theta). \quad (4-5)$$

1381 The GCE loss combines the best of both worlds, offering robustness to label noise  
1382 while maintaining the convexity property necessary for optimization. The  
1383 truncation parameter  $q$  provides a mechanism to control the sensitivity to outliers,  
1384 allowing for fine-tuning of the loss function's behavior based on the specific  
1385 characteristics of the dataset.

## 1386 4.1.2 Extension to Multiple Annotators

1387 In the context of multiple annotators, we need to consider the varying reliability  
1388 of each annotator across different regions of the image. Let's consider a  $k$ -class  
1389 multiple annotators segmentation problem with the following data representation:

$$\mathbf{X} \in \mathbb{R}^{W \times H}, \{\mathbf{Y}_r \in \{0, 1\}^{W \times H \times K}\}_{r=1}^R; \quad \mathbf{\Psi} \in [0, 1]^{W \times H \times K} = f(\mathbf{X}) \quad (4-6)$$

1390 where the segmentation mask function maps the input to output as:

$$f : \mathbb{R}^{W \times H} \rightarrow [0, 1]^{W \times H \times K} \quad (4-7)$$

1391 The segmentation masks  $\mathbf{Y}_r$  satisfy the following condition for being a softmax-like  
1392 representation:

$$\mathbf{Y}_r[w, h, :] \mathbf{1}_k^\top = 1; \quad w \in W, h \in H \quad (4-8)$$

### 1393 4.1.3 Reliability Maps and Truncated GCE

1394 The key innovation in our approach is the introduction of reliability maps  $\Lambda_r$  for  
1395 each annotator:

$$\left\{ \Lambda_r(\mathbf{X}; \theta) \in [0, 1]^{W \times H} \right\}_{r=1}^R \quad (4-9)$$

1396 These reliability maps serve as a sophisticated mechanism to estimate the  
1397 confidence of each annotator at every spatial location  $(w, h)$  in the image. By  
1398 learning these maps jointly with the segmentation model, the network gains the  
1399 ability to adapt to the varying levels of expertise across different regions of the  
1400 image. This approach allows for dynamic weighting of each annotator's

contribution based on their estimated reliability in specific areas, effectively handling cases where annotators might demonstrate different levels of expertise in different parts of the image.

The proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE<sub>SS</sub>) combines the robustness of GCE with the flexibility of reliability maps:

$$TGCE_{SS}(\mathbf{Y}_r, f(\mathbf{X}; \theta) | \Lambda_r(\mathbf{X}; \theta)) = \mathbb{E}_r \left\{ \mathbb{E}_{w,h} \left\{ \Lambda_r(\mathbf{X}; \theta) \circ \mathbb{E}_k \left\{ \mathbf{Y}_r \circ \left( \frac{\mathbf{1}_{W \times H \times K} - f(\mathbf{X}; \theta)^{\circ q}}{q} \right); k \in K \right\} + \right. \right. \\ \left. \left. (\mathbf{1}_{W \times H} - \Lambda_r(\mathbf{X}; \theta)) \circ \left( \frac{\mathbf{1}_{W \times H} - (\frac{1}{K} \mathbf{1}_{W \times H})^{\circ q}}{q} \right); w \in W, h \in H \right\}; r \in R \right\} \quad (4-10)$$

where  $q \in (0, 1)$  controls the MAE or CE level in the same way as in the GCE loss function. The loss function consists of two main components:

- The first term weighted by  $\Lambda_r$  represents the GCE loss for regions where the annotator is considered reliable
- The second term weighted by  $(1 - \Lambda_r)$  provides a uniform prior for regions where the annotator is considered unreliable

For a batch containing  $N$  samples, the total loss is computed as:

$$\mathcal{L}(\mathbf{Y}_r[n], f(\mathbf{X}[n]; \theta) | \Lambda_r(\mathbf{X}[n]; \theta)) = \frac{1}{N} \sum_n^N TGCE_{SS}(\mathbf{Y}_r[n], f(\mathbf{X}[n]; \theta) | \Lambda_r(\mathbf{X}[n]; \theta)) \quad (4-11)$$

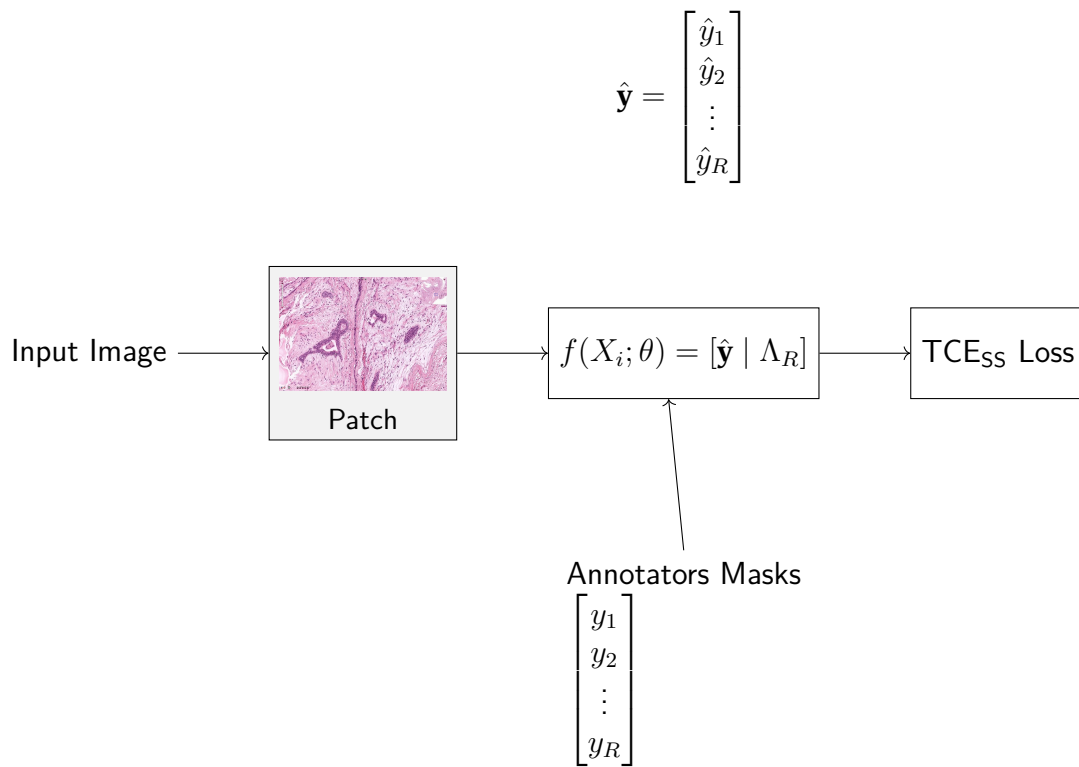


Figure 4-1 Working mechanism of the proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE<sub>SS</sub>) loss function. The loss combines reliability maps  $\Lambda_r$  with the model's predictions  $f(\mathbf{X}; \theta)$  to compute a weighted loss that accounts for annotator reliability across different image regions.



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CHAPTER

1415

**FIVE**

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CHAINED DEEP LEARNING FOR IMAGE

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SEGMENTATION

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## **5.1 Introduction**

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As mentioned in Chapter 1, U-shaped CNNs have been proven a good solution for

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segmentation tasks in medical images, due to their ability to capture both global

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and local information with relatively low datasets.

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## **5.2 Using U-Net as a Building Block**

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The proposed model architecture combines the strengths of U-Net with a ResNet-34

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backbone, specifically designed to work with the  $TGCE_{SS}$  loss function by providing

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a multi-output network which focuses on two tasks: **segmentation** and **reliability**

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**estimation**. The architecture is illustrated in Figure 5-2.

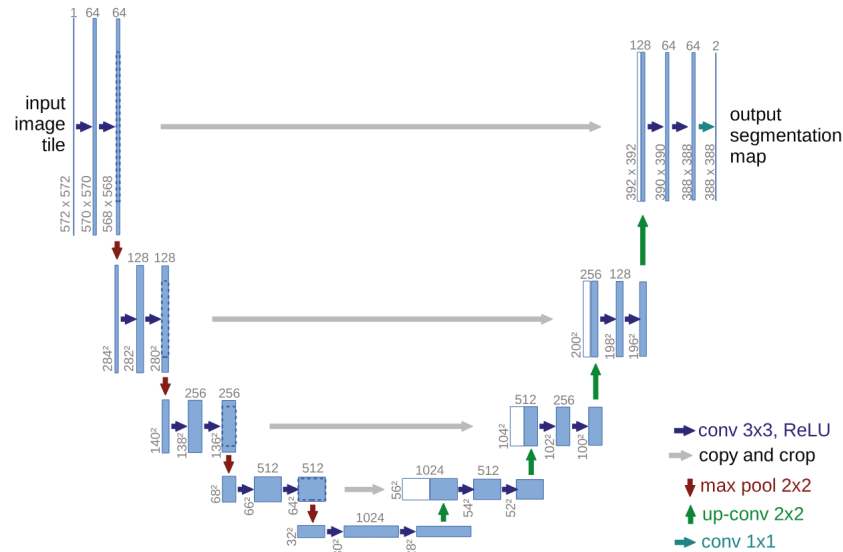


Figure 5-1 Original U-NET architecture.

## 5.2.1 Backbone Architecture

The model employs a pre-trained ResNet-34 as its encoder backbone, leveraging its deep residual learning framework for efficient feature extraction. The choice of ResNet-34 provides several key advantages: efficient feature extraction through residual connections, pre-trained weights that capture rich visual representations, and stable gradient flow during training. The ResNet-34 backbone is modified to serve as the encoder in our UNET architecture by removing the final fully connected layer and utilizing the feature maps from different stages of the network for skip connections.

## 5.2.2 U-Net Architecture

The U-Net architecture follows a traditional encoder-decoder structure with skip connections, where the encoder path implements the ResNet-34 structure. The decoder path employs transposed convolutions for upsampling, creating a

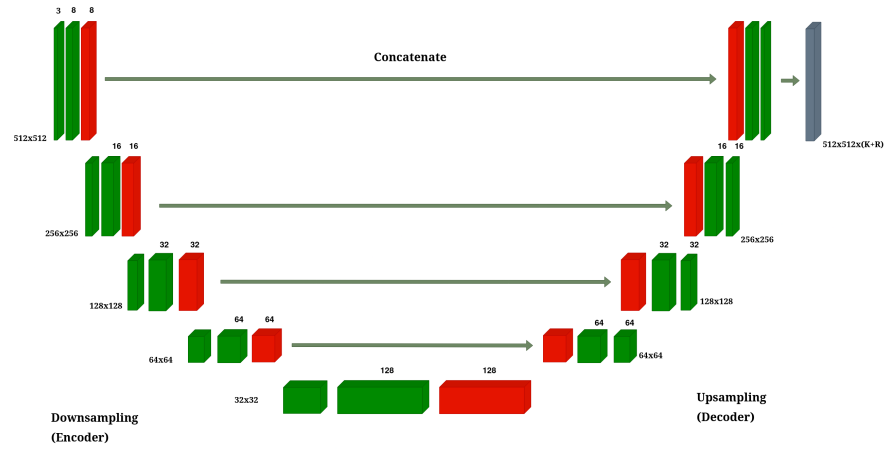


Figure 5-2 Overview of the proposed model architecture.

symmetrical architecture that effectively captures both high-level and low-level features. The architecture incorporates four downsampling stages in the encoder, corresponding to the ResNet-34 blocks, and four upsampling stages in the decoder. These stages are connected through skip connections that bridge corresponding encoder and decoder stages, allowing the network to preserve fine-grained details. Each convolution operation is followed by batch normalization and ReLU activation to ensure stable training and effective feature learning.

This U-Net architecture is expected to be coupled with a reliability branch which leads to a multi-output network. Three different approaches are considered for this branch: **scalar**, **features**, and **pixel**.

### 5.2.3 Scalar Reliability Estimation

The scalar reliability estimation branch is the simplest one, in which the reliability branch is a parallel branch to the segmentation branch, which leads to a multi-output network. The final output consists of  $R$  reliability maps  $\Lambda_r$ , one for each annotator, with values constrained to the  $[0, 1]$  range through a sigmoid activation function. This design allows the network to learn and adapt to the varying reliability of different annotators across different regions of the image.

1458 The scalar reliability branch is implemented by taking the bottleneck features from  
1459 the encoder and applying a global average pooling operation followed by a dense  
1460 layer with sigmoid activation. This results in a single reliability value per annotator  
1461 for each input image, making it computationally efficient but less spatially aware of  
1462 reliability variations within the image.

## 1463 5.2.4 Feature-based Reliability Estimation

1464 The feature-based reliability estimation approach provides a more detailed spatial  
1465 understanding of annotator reliability by operating on the bottleneck features of  
1466 the encoder. This approach maintains the spatial information from the encoder's  
1467 bottleneck while being more computationally efficient than pixel-wise reliability  
1468 estimation.

1469 The feature-based reliability branch is implemented by applying two convolutional  
1470 layers with batch normalization to the bottleneck features:

- 1471 • A first convolutional layer with 32 filters to process the bottleneck features
- 1472 • A second convolutional layer that produces  $R$  feature maps, one for each  
1473 annotator

1474 This approach provides a coarse-grained spatial reliability map that captures  
1475 regional variations in annotator reliability while maintaining computational  
1476 efficiency. The resulting reliability maps have the same spatial dimensions as the  
1477 bottleneck features, providing a middle ground between scalar and pixel-wise  
1478 reliability estimation.

### 1479 5.2.5 Pixel-wise Reliability Estimation

1480 The pixel-wise reliability estimation approach provides the most detailed spatial  
 1481 understanding of annotator reliability by operating on the full-resolution features  
 1482 from the decoder. This approach allows for fine-grained reliability estimation at the  
 1483 pixel level, capturing local variations in annotator reliability.

1484 The pixel-wise reliability branch is implemented by applying three convolutional  
 1485 layers with batch normalization to the decoder’s output features:

- 1486 • Two initial convolutional layers with 32 filters each to process the features
- 1487 • A final convolutional layer that produces  $R$  reliability maps, one for each
- 1488 annotator

1489 This approach provides the most detailed spatial reliability information, allowing  
 1490 the model to capture pixel-level variations in annotator reliability. The resulting  
 1491 reliability maps have the same spatial dimensions as the input image, providing the  
 1492 finest granularity of reliability estimation among the three approaches.

### 1493 5.2.6 Integration with $\text{TGCE}_{SS}$ Loss

1494 The model produces two distinct outputs: segmentation masks  $\hat{\mathbf{Y}} = f(\mathbf{X}; \theta)$  and  
 1495 reliability maps  $\{\Lambda_r(\mathbf{X}; \theta)\}_{r=1}^R$ . These outputs work in tandem with the  $\text{TGCE}_{SS}$   
 1496 loss function described in Section 4.1. The loss function simultaneously guides the  
 1497 learning of both the segmentation masks and reliability maps, ensuring that the  
 1498 model learns to balance the contributions of different annotators based on their  
 1499 estimated reliability.

1500 The  $\text{TGCE}_{SS}$  loss function is implemented differently for each reliability approach,  
 1501 with specific adaptations to handle the different spatial dimensions of the reliability  
 1502 maps. All implementations share common hyperparameters:

- 1503 •  $q = 0.1$ : Controls the truncation of the loss function
- 1504 •  $\epsilon = 10^{-8}$ : Small constant to prevent numerical instability
- 1505 •  $\alpha_{reg} = 0.1$ : Weight for the regularization term
- 1506 •  $\alpha_{entropy} = 0.1$ : Weight for the entropy regularization
- 1507 •  $\alpha_{sum} = 0.1$ : Weight for the sum-to-one constraint

1508 For the scalar reliability approach, the loss function operates on global reliability  
 1509 values per annotator. The reliability maps are expanded to match the spatial  
 1510 dimensions of the predictions through tiling operations. This approach is  
 1511 computationally efficient but provides the same reliability value across all spatial  
 1512 locations.

1513 The feature-based reliability approach requires additional processing to handle the  
 1514 spatial dimensions of the bottleneck features. The reliability maps are resized  
 1515 using bilinear interpolation to match the spatial dimensions of the predictions.  
 1516 This approach provides coarse-grained spatial reliability information while  
 1517 maintaining computational efficiency.

1518 The pixel-wise reliability approach operates directly on full-resolution reliability  
 1519 maps, requiring no spatial transformations. This approach provides the most  
 1520 detailed spatial reliability information but requires more computational resources.

1521 All three implementations include three regularization terms to ensure proper  
 1522 learning of the reliability maps:

- 1523 • **Center Regularization:**  $\alpha_{reg} \cdot \mathbb{E}[(\Lambda_r - 0.5)^2]$
- 1524 • **Entropy Regularization:**  $-\alpha_{entropy} \cdot \mathbb{E}[\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))]$
- 1525 • **Sum-to-One Regularization:**  $\alpha_{sum} \cdot \mathbb{E}[(\sum_r \Lambda_r - 1)^2]$

1526 These regularization terms help prevent the model from assigning extreme  
 1527 reliability values and ensure a balanced distribution of reliability across  
 1528 annotators. The loss function also includes a noise tolerance parameter ( $\eta = 0.1$ )  
 1529 to handle cases where the model's predictions differ from the annotators' labels.

1530 The final loss function combines the main  $TGCE_{SS}$  terms with the regularization  
 1531 terms:

$$\mathcal{L} = \mathbb{E}[\text{term}_1 + \text{term}_2] + \alpha_{reg} + \alpha_{entropy} + \alpha_{sum} \quad (5-1)$$

1532 where  $\text{term}_1$  and  $\text{term}_2$  are the main  $TGCE_{SS}$  terms that handle the interaction  
 1533 between predictions and reliability maps.

## 1534 5.2.7 Training Process

1535 The training process begins with the initialization of the ResNet-34 backbone  
 1536 using pre-trained weights, providing a strong foundation for feature extraction.  
 1537 The model's hyperparameters are optimized using Bayesian optimization through  
 1538 Keras Tuner, which searches for optimal values of:

- 1539 • Learning rate: Sampled logarithmically between  $10^{-5}$  and  $10^{-2}$
- 1540 •  $TGCE_{SS}$  parameters:
  - 1541 -  $q$ : Controls truncation of the loss function (0.1 to 0.9)
  - 1542 - Noise tolerance: Controls model's robustness to label noise (0.1 to 0.9)
  - 1543 - Regularization weights: Control the balance between different loss
  - 1544 terms (0.01 to 0.5)

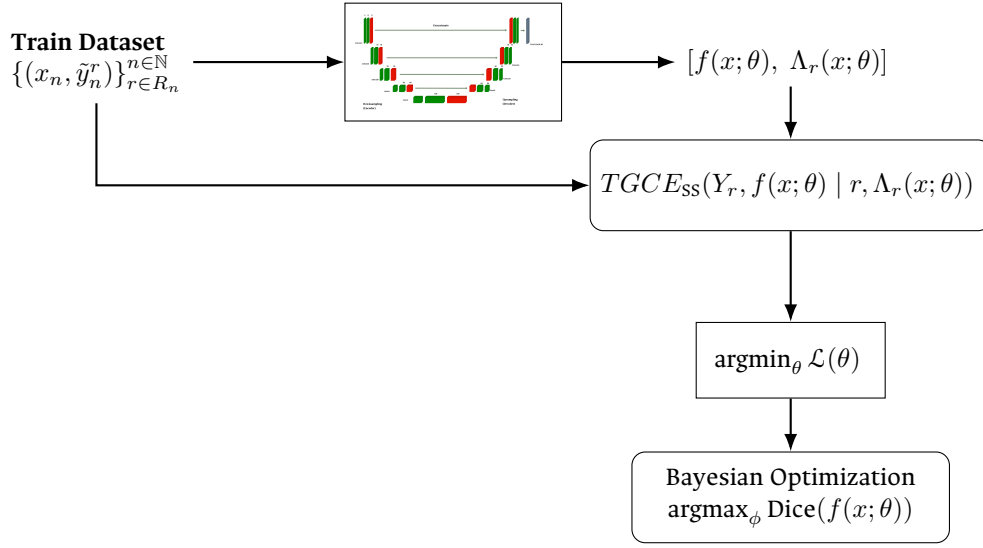


Figure 5-3 Training process of the proposed model overview. The model first undergoes hyperparameter optimization using Bayesian optimization, followed by end-to-end training with the optimal parameters. Estimated segmentation and reliability maps are computed for each input image, and the loss function is computed for the entire batch. Network parameters are tuned based on the loss function as objective (with minimization target) whereas the hyperparameters  $\phi$  tuning are based on the Dice coefficient of the validation data set (with maximization target).

1545 The hyperparameter search is performed using a validation set, with the objective  
 1546 of maximizing the Dice coefficient on the validation data. After finding the optimal  
 1547 hyperparameters, the model is trained end-to-end using the Adam optimizer with  
 1548 the best learning rate found. The  $TGCE_{SS}$  loss function plays a crucial role in  
 1549 updating both the segmentation and reliability branches, ensuring that the model  
 1550 learns to effectively handle multiple annotators' inputs while accounting for their  
 1551 varying reliability.

1552 The model's architecture is specifically designed to address the challenges of  
 1553 multi-annotator segmentation. Through the ResNet-34 backbone, it learns robust  
 1554 segmentation features that capture high-level patterns in the data. The UNET's  
 1555 skip connections enable the preservation of fine-grained details, while the parallel  
 1556 reliability branch allows the model to adapt to annotator-specific characteristics.  
 1557 This comprehensive design enables the model to effectively handle multiple



1558 annotators' inputs while maintaining high segmentation accuracy and reliability  
1559 estimation.

## 1560 5.3 Experimental Setup

### 1561 5.3.1 Noisy Annotations Generation

1562 A challenge arises for the creation of emulated noisy annotations datasets, since it  
1563 is expected for images annotations to have some degree of expertise variability,  
1564 similar to what is expected in real crowdsourced datasets. Simply introducing  
1565 random noise into the annotated masks does not work, since the original  
1566 morphological structures from the expected ground truth are far from being  
1567 preserved. Instead, a “noisy” labeler is expected to produce an annotation which  
1568 has at least some degree of morphological consistency on itself, even if it shows  
1569 discrepancies when compared with some metric (like DICE score) against the  
1570 ground truth annotation.

1571 This has been proven experimentally when introducing random noise into any  
1572 popular segmentation dataset, in which the resulting mask is just a non coherent  
1573 map of noise across the image, as shown in Figure 5-4.

1574 For this reason, a more sophisticated approach was needed to create datasets with  
1575 emulated noisy annotations. The approach used here was to use a pre-trained  
1576 U-Net model to produce a good enough <sup>1</sup> segmentation of the image, which was  
1577 then used to produce a “noisy” annotation by introducing random noise into the  
1578 last encoder layers of the model, thus preserving the morphological consistency of  
1579 the original annotation. This strategy works since slight modifications introduced

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<sup>1</sup>More precisely, a U-Net architecture with Resnet-34 backbone model with a simple categorical cross-entropy loss which reached 0.8426 Dice Coefficient after 44 epochs.

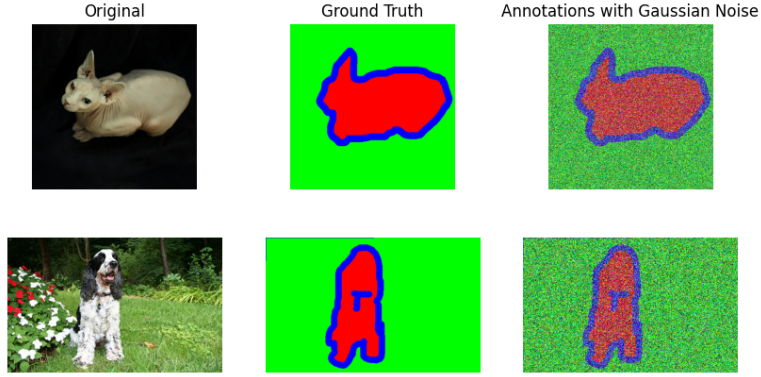


Figure 5-4 Example of a noisy mask generated by naively introducing random noise into a ground truth mask. Morphological consistency is lost.

1580 into the encoder layers weights, somehow resemble conceptual disturbances with  
 1581 respect to the respect to the original ground truth label, which kind of emulates  
 1582 the cultural behavior of having a different interpretation of the image, either for  
 1583 having a different level of expertise or for having a different point of view or  
 1584 school of thought. The first encoding layers were not modified, since it is expected  
 1585 human labelers would agree on the most fundamental structures (analog to  
 1586 extracted features from initial convolutional layers) in a similar way, thus  
 1587 preserving the morphological consistency of the original annotation.

1588 In this way, the level of “disturbance” with respect to the ground truth for an  
 1589 emulated annotator can be controlled by the level of noise introduced into the  
 1590 weights of the last encoder layers, thus:

$$\mathbf{W}_{noisy} = \mathbf{W}_{original} + \epsilon_d \quad (5-2)$$

1591 where  $\mathbf{W}_{original}$  represents the original weights of the last encoder layers,  $\epsilon_d$  follows  
 1592 a Gaussian distribution with zero mean and variance  $\sigma^2$ . The variance  $\sigma^2$  controls  
 1593 the level of noise introduced, and thus the degree of disturbance in the resulting  
 1594 segmentation masks.

With the application of the encoder layer weight perturbation technique, the resulting noisy masks are shown in Figure 5-5. It can be seen that the morphological consistency is preserved, even though the resulting masks are far from the ground truth annotations, which goes perfectly well for testing the robustness of the models against noisy annotations in crowdsourcing-like scenarios.

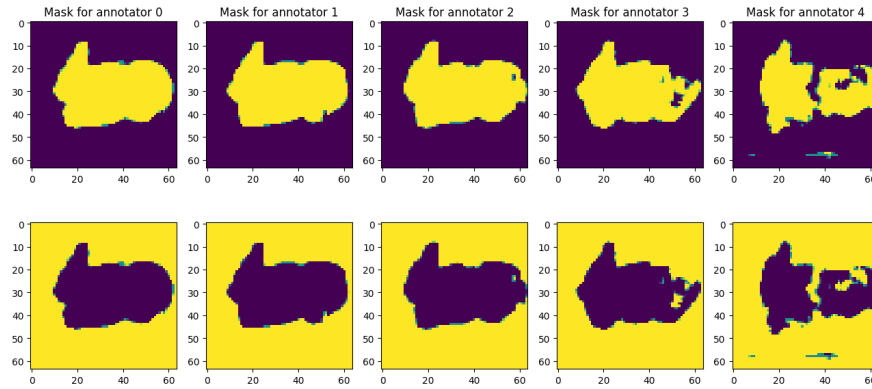


Figure 5-5 Noisy mask generated by enhancing the disturbances in the encoder layers weights for the Oxford-IIIT Pet Dataset. Morphological consistency is preserved. From left to right, SNR levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding noisy to the encoder layer would reach to an annotator behavior as good as the originally trained base model.

### 5.3.2 Missing Labels Emulation

It has also been determined that most real world crowdsourced datasets suffer from missing labels, since not all annotators are able to label all of the input samples. To emulate this behaviour, a missing labels dataset was created by randomly skipping labels from the annotators labels based on a labeling rate  $r_l \in [0, 1]$ . The labeling rate  $r_l$  controls the proportion of labels that are skipped from the annotators labels. The lower the labeling rate, the more labels are skipped from the annotators labels. The implementation of this missing labels emulation together with the noisy annotators generation can be flexibilized through an algorithm, which enables it to be used in

1610 experiments with variable amount of labelers, noise levels, and labeling rates. An  
 1611 overview of the algorithm is shown in Algorithm 5-1. A sample of the generated  
 1612 noisy and missing labels dataset is shown in Figure 5-6.

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**Algorithm 5-1** Multi-Annotator Dataset Generation with Noisy and Missing Labels

---

**Require:** Base segmentation model  $M$ , SNR levels  $\{s_1, \dots, s_n\}$ , labeling rate  $r \in [0, 1]$

**Ensure:** Dataset with multiple noisy annotations and missing labels

```

1: Step 1: Generate Noisy Annotators
2: for each SNR level  $s_i$  do
3:   Create disturbed model  $M_i$  by adding noise to  $M$  with SNR  $s_i$ 
4: end for
5: Step 2: Initialize Labeler Assignment System
6: Create assignment matrix  $A \in \{0, 1\}^{N \times n}$  where  $N$  is number of samples
7: for each annotator  $j \in \{1, \dots, n\}$  do
8:   for each sample  $i \in \{1, \dots, N\}$  do
9:      $A_{i,j} \leftarrow \begin{cases} 1 & \text{with probability } r \\ 0 & \text{with probability } 1 - r \end{cases}$ 
10:   end for
11: end for
12: Step 3: Generate Multi-Annotator Dataset
13: for each image  $x$  in dataset do
14:   Resize  $x$  to target dimensions
15:   for each annotator model  $M_i$  do
16:     Generate annotation  $y_i = M_i(x)$ 
17:     Apply labeler assignment mask:  $y_i \leftarrow y_i \cdot A_{i,j}$ 
18:   end for
19:   Store  $(x, \{y_1, \dots, y_n\})$  in dataset
20: end for
   return Multi-annotator dataset with noisy and missing labels

```

---

| Method       | Label Rate 1.0  |                 | Label Rate 0.7      |                     | Label Rate 0.3      |                     |
|--------------|-----------------|-----------------|---------------------|---------------------|---------------------|---------------------|
|              | Dice            | Jaccard         | Dice                | Jaccard             | Dice                | Jaccard             |
| TGCE scalar  | 0.8622±0.009    | 0.7791±0.012    | <b>0.8568±0.017</b> | <b>0.7726±0.024</b> | <b>0.8529±0.004</b> | <b>0.7677±0.007</b> |
| TGCE feature | 0.6995±0.044    | 0.5733±0.047    | 0.4766±0.026        | 0.3441±0.027        | 0.4766±0.026        | 0.3441±0.027        |
| TGCE pixel   | 0.8468±0.018    | 0.7589±0.025    | 0.8545±0.013        | 0.7695±0.017        | 0.8283±0.001        | 0.7359±0.002        |
| MV           | 0.8132±0        | 0.7222±0        | 0.7572±0            | 0.6554±0            | 0.5566±0            | 0.4687±0            |
| STAPLE       | <b>0.8718±0</b> | <b>0.7917±0</b> | 0.8439±0            | 0.7609±0            | 0.5734±0            | 0.5158±0            |

Table 5-1 Performance metrics (Dice and Jaccard) at different labeling rates. Best values per category are bolded.

## 5.4 Results

### 5.4.1 Oxford-IIIT Pet Dataset

The experimental results on the Oxford-IIIT Pet Dataset demonstrate interesting insights into the performance of different approaches. While STAPLE achieves the highest Dice and Jaccard scores (0.8718 and 0.7917 respectively) for the dataset with no missing labels (label rate of 1.0), our proposed TGCE<sub>SS</sub> approaches show superior performance for label rates below 1.0 (Table 5-1). The pixel-wise variant, while slightly lower in performance metrics, provides more detailed spatial reliability information. It is of note that the key strength of TGCE<sub>SS</sub> lies in its ability to simultaneously learn both segmentation and reliability estimation, providing valuable insights into annotator behavior and confidence levels that STAPLE and MV cannot offer.

The reliability estimation capabilities of our model are particularly noteworthy, as demonstrated in Figures 5-7 and 5-8. The model successfully captures the varying reliability levels of different annotators, with higher reliability assigned to annotators with lower noise levels (10 dB) and lower reliability to those with higher noise levels (-20 dB). This ability to quantify and visualize annotator reliability is a significant advantage over traditional approaches like STAPLE and MV, which focus solely on segmentation accuracy without providing insights into annotator behavior.

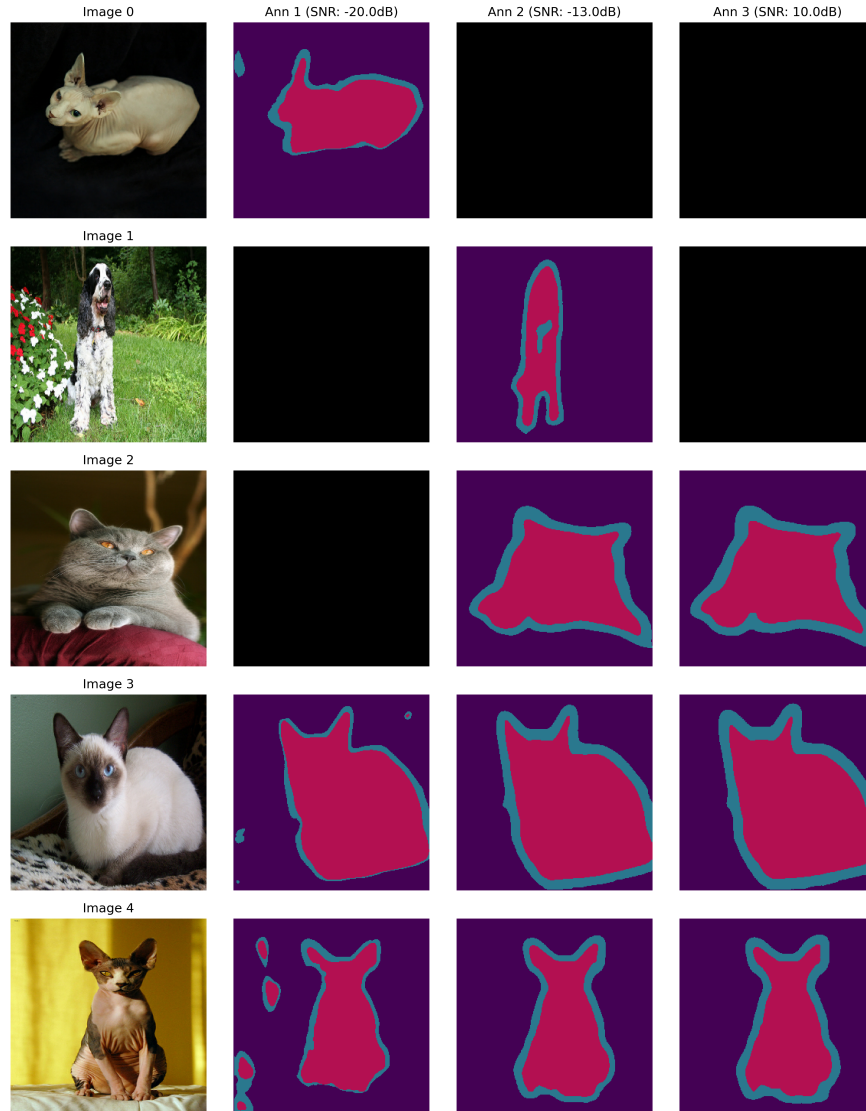


Figure 5-6 Example of missing labels and noisy annotations emulation for the Oxford-IIIT Pet Dataset. The same label rate ( $r_l = 0.7$ ) is preserved for all labelers, which means in this case that every labeler is expected to label 70% of the samples, and skip the remaining 30%.

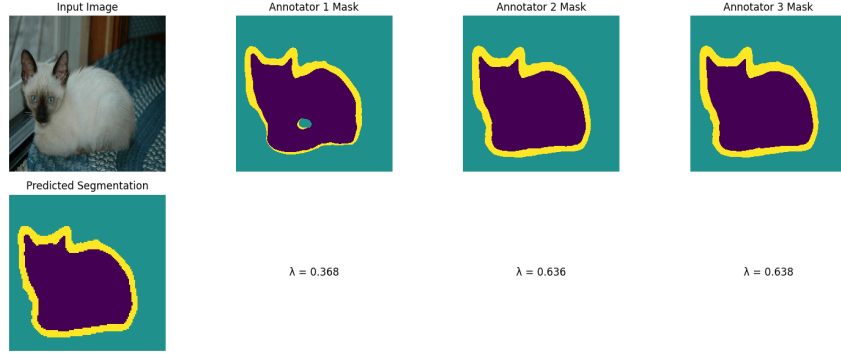


Figure 5-7 Example evaluation of the proposed model in its last trained epoch. An accurate segmentation output is delivered as well as scalar reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are:  $[-20, 0, 10]$  dB from left to right. Best found hyperparameters: learning rate = 0.0011,  $q = 0.4$ , noise tolerance = 0.2,  $\alpha_{reg} = 0.26$ ,  $\alpha_{entropy} = 0.03$ ,  $\alpha_{sum} = 0.27$ .

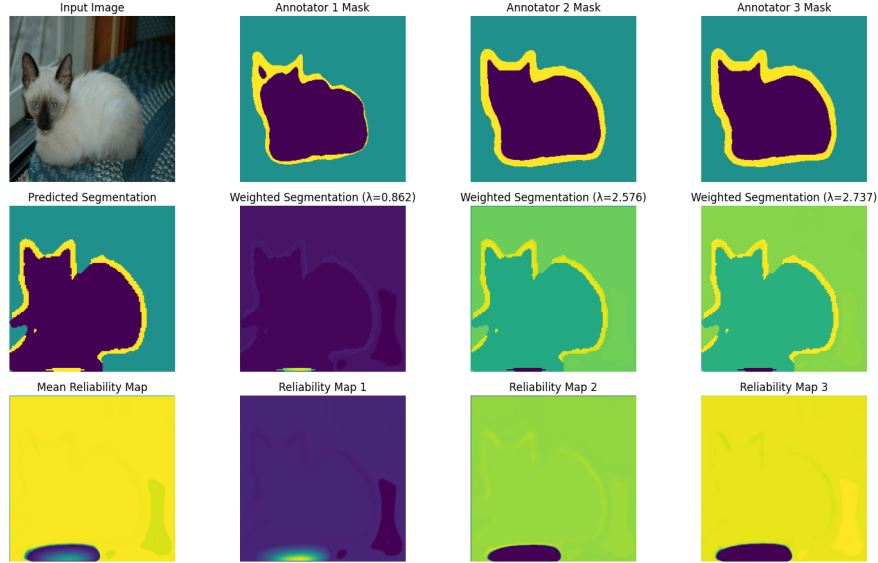


Figure 5-8 Example evaluation of the proposed model in some given epoch (5/50) of training, which demonstrates a quick convergence of the model for correctly estimating the reliability of the labelers masks. An accurate segmentation output is delivered as well as pixel-wise reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are:  $[-20, 0, 10]$  dB from left to right. Best found hyperparameters: learning rate = 0.0027,  $q = 0.9$ , noise tolerance = 0.2,  $\alpha_{reg} = 0.38$ ,  $\alpha_{entropy} = 0.02$ ,  $\alpha_{sum} = 0.42$ .





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1634

CHAPTER

1635

**SIX**

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FINAL REMARKS

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## 6.1 Conclusions

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Throughout this work, several techniques have been explored to develop a model that can face **ISS** of real world histopathology images in crowdsourcing scenarios.

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Different approaches were studied and evaluated to reach an optimal solution

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which could model the annotators' performance from the input space. The

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following remarks are based on the results obtained from the experiments

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conducted and reviewed literature:

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- Chained Gaussian Processes approaches, while robust for classification and regression tasks, proved computationally intensive for image segmentation. This limitation highlighted the need for more efficient approaches that could handle the complexity of medical image segmentation while modeling annotator performance.

- The development of a novel loss function was crucial for the success of this work, as existing loss functions were insufficient to model the annotators' performance across the input space. The proposed loss function successfully enabled the model to adaptively infer the best possible segmentation without requiring separate inputs about labeler performance.
- The introduction of a tensor map to codify labeler reliability proved to be an effective approach, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space, thus handling the labeler performance inconsistency effectively.
- The use of U-Shaped Deep Learning models as a building block for the proposed solution was validated, as it effectively captured spatial relationships between pixels and annotations while handling the variability in image quality.
- The combination of Chained Deep Learning approaches with the proposed loss function and tensor map representation resulted in a comprehensive solution being, to the best of our knowledge, the first approach on offering an image segmentation solution for multiple labelers in medical image with interpretability in the reliability of the labelers, successfully addressing both labeler inconsistency and image quality variability.

## 6.2 Future work

As tools and frameworks for bayesian approaches like Chained Gaussian Processes<sup>1</sup> evolve, it is expected to see a decrease in computational requirements and hence, making mixed models more appealing for the task of ISS of histopathology images. In that sense, the use of CCGPMA as a building block for the proposed solution could

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<sup>1</sup>Including GPFlow, and GPFlux.

be a good approach, as it is a powerful model that can take advantage of the inter-dependencies between the annotators, and the powerfulness of CNNs to extract features from the images.

The noisy labelers emulation technique developed in this work presents an opportunity to test the model's robustness in various medical image segmentation challenges. This technique could be adapted to simulate different types of expert variability in other medical imaging domains:

- **Multi-modal Medical Imaging:** The technique could be extended to simulate expert disagreements in multi-modal scenarios, where different imaging modalities (e.g., MRI, CT, PET) might lead to varying interpretations of the same anatomical structure.
- **Temporal Consistency:** For longitudinal studies, the emulation could be modified to simulate how expert annotations might vary over time, helping to test the model's ability to maintain consistency in segmentation across different time points.
- **Cross-institutional Variability:** The noise injection technique could be calibrated to simulate the variations in annotation patterns that arise from different institutional protocols or training backgrounds.
- **Pathology-specific Challenges:** The emulation could be tailored to simulate expert disagreements in specific pathological conditions, where the morphological features might be more ambiguous or subject to interpretation.

These extensions would help validate the model's performance across a broader range of medical imaging scenarios and ensure its robustness in real-world clinical applications.

The pixel-wise reliability mapping approach shows particular promise for future development. While the current implementation already demonstrates good performance, several enhancements could lead to significant improvements:

1702 • **Advanced Data Augmentation:** Implementing a comprehensive data  
1703 augmentation strategy could significantly improve the model's convergence  
1704 and performance. This could include:

- 1705       – Geometric transformations (rotations, flips, elastic deformations)
- 1706       – Intensity variations to simulate different staining conditions
- 1707       – Multi-scale training to handle varying tissue sizes
- 1708       – Adversarial training to improve robustness

1709 These augmentations would help the model learn more robust features and  
1710 better generalize to unseen data.

1711 • **Attention Mechanisms:** Integrating attention mechanisms into the pixel-wise  
1712 reliability branch could help the model focus on the most relevant regions for  
1713 reliability estimation, potentially improving both accuracy and computational  
1714 efficiency.

1715 • **Multi-scale Reliability Estimation:** Implementing a hierarchical approach to  
1716 reliability estimation could capture both fine-grained local variations and  
1717 broader regional patterns in annotator reliability, this would imply a  
1718 network with yet more branches and outputs.

1719 The combination of these enhancements with the existing pixel-wise reliability  
1720 mapping approach could potentially outperform all other techniques, as it would  
1721 provide the most detailed and accurate representation of annotator reliability  
1722 while maintaining computational efficiency. The use of data augmentation, in  
1723 particular, could help the model converge faster and achieve better generalization,  
1724 as it would expose the model to a wider variety of possible variations in both the  
1725 input images and the corresponding annotations.

## 6.3 Academic Discussion

To facilitate the adoption and further development of the proposed approach, several resources have been made publicly available. A comprehensive GitHub repository has been created that provides easy-to-use APIs for implementing the proposed solution. The repository includes pre-built dataset loaders for TensorFlow/Keras frameworks, implementation of the  $TGCE_{SS}$  loss function in all its variants (scalar, features, and pixel-wise), and a neural network architecture builder with experimental examples. A key feature of this repository is its automatic dataset fetching from AWS S3, which eliminates the need for manual downloads or external dependencies. This repository aims to lower the entry barrier for researchers and practitioners interested in implementing the proposed approach.

To further simplify the adoption of the proposed solution, a PyPI package has been published with detailed documentation available at <https://seg-tgce.readthedocs.io/en/latest/>. The documentation provides comprehensive installation instructions, API reference, usage examples, and best practices for implementation. This package makes it straightforward for the community to integrate the proposed approach into their existing workflows, with a focus on ease of use and clear documentation of all available features and options.

An academic publication is currently being prepared for submission to a Q1 journal. This publication will provide detailed theoretical foundations of the proposed approach, comprehensive experimental results and analysis, comparison with state-of-the-art methods, and discussion of limitations and future directions. The publication will serve as the primary reference for the theoretical and practical aspects of the proposed solution, ensuring that the work is properly documented and validated through the peer-review process.

These resources collectively contribute to the reproducibility and accessibility of the research, enabling the broader scientific community to build upon and extend

1755 the proposed approach. The combination of open-source code, well-documented  
1756 APIs, and peer-reviewed publication ensures that the work can be properly  
1757 evaluated, validated, and utilized by other researchers in the field. This  
1758 comprehensive approach to sharing research outputs aligns with current best  
1759 practices in scientific computing and machine learning research, promoting both  
1760 transparency and practical utility of the proposed solution.

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